# $\chi_2\big(\sqrt{M^2+1}-M\big)$ IS IRRATIONAL FOR EVERY $M \geq 16$

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ABSTRACT. We prove that the value $\chi_2(\sqrt{m^2+1}-m)$ of Legendre's chi function is irrational for every integer $m \geq 2$ except $m = 3, 5, 7, 9, 15$, that is, for $m = 2, 4, 6, 8$, $10 \leq m \leq 14$ and every $m \geq 16$. For $m = 2$ this says that $\pi^2 - 18\log^2\varphi$ is irrational, where $\varphi$ is the golden ratio, since $\chi_2(\sqrt{5}-2) = \pi^2/24 - (3/4)\log^2\varphi$. The stronger statement $\chi_2(\sqrt{m^2+1}-m) \notin \mathbb{Q}(\sqrt{m^2+1})$ is already due to David, Hirata-Kohno and Kawashima (2022) for every $m \geq 4\,346\,237$, through their linear independence criterion; the new range is $2 \leq m \leq 4\,346\,236$, $m \notin \{3, 5, 7, 9, 15\}$, where we know of no earlier result. The point $z_m = \sqrt{m^2+1} - m$ is a unit of norm $-1$ in $\mathbb{Q}(\sqrt{m^2+1})$. Other cases are $\chi_2(\sqrt{17}-4)$, $\chi_2(\sqrt{37}-6)$, $\chi_2(\sqrt{65}-8) \notin \mathbb{Q}$; for $m \geq 3$ these values have no Landen-type closed form (we show this at the level of symbols). We take poles at the even integers $\pm 2j$, $|j| \leq h/2$, and atoms at the odd positive integers. A reflection $t \mapsto -t$, combined with the Galois conjugation $z \mapsto -1/z$, shows that the Hankel determinants are values of polynomials with rational coefficients, so the conjugate place, where the series diverges, costs nothing. The odd primes not dividing $m$ cost at most $4/3$ per $h^2$; the odd primes dividing $m$ and the prime 2 cost explicit amounts that depend on $m$. The size is bounded by a discrete log-gas argument against a dual certificate checked in interval arithmetic at $m = 2$, and a shift lemma carries this bound to every $m$. Nonvanishing comes from a reduction modulo a prime $\ell = (h+1)/3$. The theorem, for every $m$, is formalised in Lean 4 with Mathlib, using only Lean's three standard axioms. We also prove a two-point statement at units of a family of biquadratic fields: for $n \in \{730, 740, 750, 760, 800, 1000\}$, either $\chi_2(z) + \chi_2(w) \notin \mathbb{Q}$ or $\chi_2(z) - \chi_2(w) \notin \sqrt{n^2-16}\,\mathbb{Q}$, where $z > w$ are the two roots in $(0,1)$ of $X^4 + nX^3 + 2X^2 - nX + 1$. This second result is computer-assisted and not formalised.

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## 1. Introduction

For $|q| \leq 1$ let

$$\chi_2(q) = \sum_{n=0}^{\infty} \frac{q^{2n+1}}{(2n+1)^2} = \tfrac{1}{2}\big(\mathrm{Li}_2(q) - \mathrm{Li}_2(-q)\big)$$

be Legendre's chi function [18, §1.8]. For an integer $m \geq 1$ put

$$z_m := \sqrt{m^2+1} - m \in (0,1).$$

Then $z_m^2 + 2mz_m - 1 = 0$, so $z_m$ is a unit of norm $-1$ in the real quadratic field $\mathbb{Q}(\sqrt{m^2+1})$, with Galois conjugate $-1/z_m$. Put $\mathcal{E} := \{3,5,7,9,15\}$.

**Theorem 1.1** (Theorem V)**.** *For every integer $m \geq 2$ with $m \notin \mathcal{E}$, the number $\chi_2(\sqrt{m^2+1}-m)$ is irrational.*

For $m \geq 4\,346\,237$ the stronger statement $\chi_2(z_m) \notin \mathbb{Q}(\sqrt{m^2+1})$ is due to David, Hirata-Kohno and Kawashima [4, Thm. 1] (Section 8.2). What Theorem 1.1 adds is the range $2 \leq m \leq 4\,346\,236$ with $m \notin \mathcal{E}$, where we know of no earlier result. The proof is computer-assisted: it uses one dual certificate checked in interval arithmetic (at $m = 2$) and a table of margins for the odd $m \leq 65$, also computed in interval arithmetic. The case $m = 2$ is the golden-ratio statement. Let $\varphi = (1+\sqrt{5})/2$.

**Corollary 1.2** (Theorem U)**.** *The number*

$$\pi^2 - 18\log^2\varphi = 5.70143763069985955388\ldots$$

*is irrational. Equivalently, the number*

$$\chi_2(\sqrt{5}-2) = \frac{\pi^2}{24} - \frac{3}{4}\log^2\varphi = 0.23755990127916081474\ldots$$

*is irrational.*

The two forms are equivalent by the identity

$$\chi_2(\sqrt{5}-2) = \frac{\pi^2}{24} - \frac{3}{4}\log^2\varphi, \tag{1.1}$$

which follows from Landen's identity $\chi_2\big(\frac{1-x}{1+x}\big) + \chi_2(x) = \frac{\pi^2}{8} + \frac{1}{2}\log x \,\log\frac{1+x}{1-x}$ [18, App. A.2.1, (24)] at $x = \sqrt{5}-2 = \varphi^{-3}$, for which $(1-x)/(1+x) = \varphi^{-1}$, together with the classical value $\chi_2(\varphi^{-1}) = \pi^2/12 - \frac{3}{4}\log^2\varphi$. Theorem 1.1, Corollary 1.2 and the identity (1.1) are formalised in Lean (Section 10).

Theorem 1.1 covers every even $m \geq 2$ and every odd $m \geq 11$ except 15. For example,

$$\chi_2(\sqrt{17}-4) \notin \mathbb{Q}, \qquad \chi_2(\sqrt{37}-6) \notin \mathbb{Q}, \qquad \chi_2(\sqrt{65}-8) \notin \mathbb{Q}.$$

For $m \geq 3$ these values are of a different kind from $\chi_2(\sqrt{5}-2)$: they have no expression through $\pi^2$ and products of logarithms by dilogarithm identities that preserve the symbol (Proposition 8.1). Only $m = 1$ and $m = 2$ have a Landen closed form. The method does not reach $m = 1$, where $\chi_2(\sqrt{2}-1) = \pi^2/16 - \frac{1}{4}\log^2(1+\sqrt{2})$, nor $m \in \mathcal{E}$ (Section 7.2).

1.1. **Context.** *General $m$.* For $m \geq 4\,346\,237$, Theorem 1 of David, Hirata-Kohno and Kawashima [4] gives $\chi_2(z_m) \notin \mathbb{Q}(\sqrt{m^2+1})$, which is stronger than Theorem 1.1; this is our evaluation of their criterion (Section 8.2). So the infinite part of Theorem 1.1 is not new by itself. We have found no result that covers any $m$ with $2 \leq m \leq 4\,346\,236$. For $m \geq 3$ that search was short: our own notes, two web searches, and a reading of the criteria listed in Section 8.2; it did not cover MathSciNet, and no expert has been consulted. Our method also covers the range of [4] independently, with the weaker conclusion.

*The case $m = 2$.* We have not found Corollary 1.2, or the irrationality of any number $\pi^2 + c\log^2\varphi$ with $c \in \mathbb{Q}\setminus\{0\}$, in the literature, and as far as we can see it does not follow from known results. By the analytic class number formula, $L(1,\chi_5) = 2\log\varphi/\sqrt{5}$ for the quadratic character $\chi_5$ modulo 5, so the corollary can also be written as $6\zeta(2) - \frac{45}{2}L(1,\chi_5)^2 \notin \mathbb{Q}$.

- *The golden-ratio dilogarithm values are different numbers.* The classical evaluations
$$\operatorname{Li}_2(\varphi^{-2}) = \tfrac{\pi^2}{15} - \log^2\varphi, \qquad \operatorname{Li}_2(\varphi^{-1}) = \tfrac{\pi^2}{10} - \log^2\varphi,$$
$$\operatorname{Li}_2(-\varphi^{-1}) = -\tfrac{\pi^2}{15} + \tfrac{1}{2}\log^2\varphi, \qquad \operatorname{Li}_2(-\varphi) = -\tfrac{\pi^2}{10} - \log^2\varphi$$
(Landen; see [18, 19]) all lie in $\mathbb{Q}\pi^2 + \mathbb{Q}\log^2\varphi$. Normalised to the form $\pi^2 + c\log^2\varphi$ they have $c = -15$, $-10$, $-\frac{15}{2}$ and $+10$, while Corollary 1.2 has $c = -18$. Since $\log\varphi$ is transcendental, $\pi^2 + c\log^2\varphi$ can be algebraic for at most one $c \in \overline{\mathbb{Q}}$; this does not say which. The irrationality of one member of the family says nothing about another, and as far as we know the irrationality of each of the four classical values is open.
- *Homogeneous statements.* The theorem of Gel'fond and Schneider [1, Ch. 2], applied to $e^{\log\varphi} = \varphi$ and $e^{i\pi} = -1$, shows that $\log\varphi/(i\pi)$ is transcendental, so $\log^2\varphi/\pi^2 \notin \overline{\mathbb{Q}}$. This says that $\pi^2$ and $\log^2\varphi$ are linearly independent over $\mathbb{Q}$; it says nothing about a relation that also involves 1. The only member of the family $\pi^2 + c\log^2\varphi$, $c \in \mathbb{Q}$, whose irrationality we know to be established is $c = 0$. The linear independence of $1, \pi^2, \log^2\varphi$ over $\mathbb{Q}$ is open as far as we know.
- *Criteria at algebraic points.* The linear independence criteria for polylogarithms and Lerch functions at algebraic points [3, 5, 4, 13],

and the explicit form of the Bombieri–André principle of global relations [21], need a point that is small in comparison with its height. Here $\chi_2(\sqrt{5}-2) = \frac{1}{2}(\mathrm{Li}_2(z) - \mathrm{Li}_2(-z)) = z + \frac{z}{4}\Phi_2(\frac{1}{2}, z^2)$, where $\Phi_2(x,q) = \sum_{k\geq 0} q^{k+1}/(k+x+1)^2$ is the Lerch function and $z = \sqrt{5}-2 = \varphi^{-3}$ is a unit of height $\frac{3}{2}\log\varphi = 0.7218\ldots$ whose conjugate $-\varphi^3$ has absolute value $> 1$, so that $|z|\,|\sigma z| = 1$. Evaluating the criteria in the form they are stated, their key quantity is negative by a wide margin at this point: about $-26.1$ for [13, Thm. 2.2], and about $-15.1$ and $-14.5$ for the two ways of applying [4, Thm. 1] (through $\mathrm{Li}_2(\pm z)$, or through $\Phi_2(\frac{1}{2}, z^2)$). The Bombieri–André bound of [21], applied to this system, needs by our estimate a height of about 1450. These evaluations are ours and have not been checked by the authors of the criteria.

- *Rational points.* The irrationality and linear independence results for dilogarithms at rational points [9, 10, 23, 24, 31], the recent claims [22, 26] (unrefereed), and our result $\chi_2(1/3), \chi_2(1/5) \notin \mathbb{Q}$ [15] all concern rational arguments, as do the weight-two results of Calegari, Dimitrov and Tang [2] (at sixth roots of unity, and for products $\log(1+1/m)\log(1+1/n)$) and Hata's irrationality of $\log(1+1/q)\log(1-1/q)$ [11]. The results on products of logarithms are homogeneous and contain no $\pi^2$.
- *The closest analogue.* The nearest result we know is $\mathrm{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2}\log^2 2 \notin \mathbb{Q}$, proved in Lean by Q. Tang [27] and, with many more rational points, by S. Liu [20]; neither is accompanied by a refereed paper. It has the same shape, $\pi^2 + c\log^2$ of a number, with $c = -6$ and the number 2 instead of a unit. Over $\mathbb{Q}(\sqrt{5})$ we have found nothing comparable.

For $m = 2$ our searches for prior work, in four passes, covered the web, arXiv (through its API), zbMATH Open, OpenAlex, Crossref, Semantic Scholar (only partly, because of rate limits), the Hexagon repository, the `openai/math` manuscript list and recent Lean repositories (the last pass on 7 November 2026). They did not cover MathSciNet or Google Scholar, and no expert has been consulted. All searches, for $m = 2$ and for general $m$, were carried out by AI systems.

1.2. **The method.** Fix $m \geq 1$ and put

$$z := z_m = \sqrt{m^2+1} - m, \qquad y := z^2, \qquad T := \chi_2(z) = \sum_{t \text{ odd}\geq 1} \frac{z^t}{t^2}.$$

Then $z^2 + 2mz - 1 = 0$, so $1 - y = 2mz$, and the Galois conjugate of $z$ is $-1/z$. For $m = 2$ this is $z = \sqrt{5}-2$ and $1-y = 4z$. The proof transposes the proof of [15] for $\chi_2(1/q)$ to these points. For even $h$ we take the $K := h+1$ poles $x_k := 2(k - h/2)$, i.e. the even integers in $[-h, h]$, put $D(t) := \prod_k (t - x_k)$,

and consider the functional

$$\Lambda(g) := \sum_{t \text{ odd} \geq 1} z^t \Big(\frac{g}{D}\Big)'(t) \qquad (g \in \mathbb{Q}[t]).$$

Partial fractions give $\Lambda = A + T\,B$ with $\mathbb{K}$-valued functionals $A, B$, where $\mathbb{K} := \mathbb{Q}(\sqrt{m^2+1})$, so the Hankel determinant $\det(\Lambda(t^{i+j}))_{i,j<h}$ is the value at $X = T$ of $\Delta_h(X) := \det(A(t^{i+j}) + XB(t^{i+j}))_{i,j<h}$, a polynomial of degree at most $h$.

What differs from [15] is the *reflection* (Lemma 2.2). The pole set is symmetric under $t \mapsto -t$, $D(-t) = -D(t)$, and the conjugation $\sigma\colon z \mapsto -1/z$ maps the partial sums of $\chi_2$ at the poles $x$ to those at $-x$. Together these give $\sigma(\Lambda_X(t^a)) = (-1)^a \Lambda_X(t^a)$, which forces $\Delta_h \in \mathbb{Q}[X]$ although its entries lie in $\mathbb{K}$. So we never pay for the conjugate place, where $|\sigma z| = 1/z > 1$ and the defining series diverges. The three inputs are then as in [15].

(1) *Denominators* (Sections 3–4). The moments of $\Lambda$ have the denominators $(1-y)^{b+1} = (2mz)^{b+1}$. At odd primes $p \nmid m$ the elements $z$, $y$, $1-y$ are units, and the bounds do not depend on $m$. Primes $p < K/3$ obey a plateau bound $2h\,\mathrm{lev}^*(p) + v_p(h!)$, which we prove through a bordered version of the Lagrange Gram identity (Lemmas 3.4 and 3.5), and primes $K/3 \leq p < K$ obey a class count $3(K-p)$ (Lemmas 3.2 and 3.3). The prime number theorem sums these to $\frac{2}{3} + \frac{2}{3} = \frac{4}{3}$ per $h^2$ (Theorem 3.7). An odd prime $p$ with $e := v_p(m) \geq 1$ costs at most $c_p(e) := \big(\frac{e}{4} + \frac{1}{4(p-1)}\big)\log p$ per $h^2$ (Lemma 3.8). The prime 2 costs at most $c_2(\nu) := |\nu - 2|\frac{\log 2}{4}$ per $h^2$, where $\nu := 1 + v_2(m)$ is the 2-adic valuation of $1-y$ (Lemma 4.1); for $m = 2$, $\nu = 2$ and the cost is only $O(h \log h)$ (Lemma 4.3).

(2) *Size* (Section 5). A discrete log-gas argument bounds $h^{-2}\log|\Delta_h(T)|$ by the value $\Phi_z$ of any dual certificate for a constrained equilibrium problem with external field $x\log z - \frac{1}{2}\int_{x-1}^{x+1}\log|s|\,ds$ (Theorem 5.1). An explicit certificate at $z = \sqrt{5}-2$, checked in interval arithmetic, gives $\Phi \leq \Phi_U := -1.3592161834586374$ (Theorem 5.3). A shift lemma (Lemma 5.5) moves it to $z_m$ and gives $\Phi \leq \Phi_U - \frac{1}{2}L(m)$ with $L(m) := \log\big((m+\sqrt{m^2+1})/(2+\sqrt{5})\big)$ (Corollary 5.6).

(3) *Nonvanishing* (Section 6). For $h = 3\ell - 1$ with $\ell \geq 7$ prime and $\ell \nmid m$, the Gram matrix of $\Lambda$ in the Lagrange basis is block diagonal modulo a prime of $\mathbb{K}$ above $\ell$, with $3\times 3$ blocks in closed form, so $\Delta_h(a/b) \neq 0$ whenever $\ell^2 \nmid b$ (Lemma 6.1).

The proof of Theorem 1.1 therefore shows

(1.2)
$$\limsup_h \frac{1}{h^2}\log|F_h(T)| \leq M(m) := \Phi_U - \tfrac{1}{2}L(m) + \tfrac{4}{3} + c_2(\nu) + \sum_{p^e \| m,\ p \text{ odd}} c_p(e)$$

along a suitable infinite set of $h$, for primitive $F_h \in \mathbb{Z}[X]$ of degree $\leq h$ with $F_h(T) = \Delta_h(T) \cdot (\text{integer})$, and $M(m) < 0$ exactly when $m \geq 2$ and $m \notin \mathcal{E}$ (Section 7.2). Table 1 shows the budget for $m = 2$ and $m = 4$.

| contribution (per $h^2$, lim sup) | source | $m = 2$ | $m = 4$ |
|---|---|---|---|
| odd primes $p \nmid m$ | Thm. 3.7 | 1.333333 | 1.333333 |
| $p = 2$ | Lemmas 4.3, 4.1 | 0 | 0.173287 |
| odd primes $p \mid m$ | Lemma 3.8 | none | none |
| arithmetic | | 1.333333 | 1.506620 |
| analytic $\Phi$ | Cor. 5.6 | $\leq -1.359216$ | $\leq -1.684755$ |
| margin $M(m)$ | | $\leq -0.025882$ | $\leq -0.178135$ |

TABLE 1. The exponential budget for $m = 2$ (Corollary 1.2) and $m = 4$ ($\chi_2(\sqrt{17}-4)$). For $m = 2$ the analytic value is the upper endpoint $-1.3592161834586374$ of the certified Arb ball, rounded up, and the margin is $-0.0258828501\ldots$, rounded up. Other values of $m$ are in Table 4.

For $m = 2$ the margin is about 1.9% of the arithmetic part. It is smaller than in [15] for $q = 3$ in relative terms, but no constant in the budget is estimated: each is a proved upper bound. The margins vary with $m$; among $m \notin \mathcal{E}$ the smallest in absolute value are $m = 11$ ($-0.0179$), $m = 21$ ($-0.0204$) and $m = 2$ ($-0.0259$).

1.3. **The formalisation.** Theorem 1.1, Corollary 1.2 and the identity (1.1) are formalised in Lean 4 [6] with Mathlib [29] and checked by the Lean kernel. The headline theorems are `Chi2U.V.theoremV` (Theorem 1.1), `Chi2U.chi2_sqrt5_sub_two_irrational` ($\chi_2(\sqrt{5}-2) \notin \mathbb{Q}$), `Chi2U.golden_irrational'` ($\pi^2 - 18\log^2\varphi \notin \mathbb{Q}$) and `Chi2U.chi2_sqrt5_sub_two_eq` (the identity (1.1)). They depend only on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`); there is no `sorry` and no `native_decide`. The formal proof of the analytic bound follows the lattice-energy route of [15], not the route of Section 5, and uses its own, coarser certificate; Section 10 describes both differences. For general $m$ the formal proof transports this certificate by the shift lemma; the exception set is the same $\mathcal{E}$.

1.4. **Two points.** Section 9 transposes the proof once more, to units of the biquadratic fields $\mathbb{Q}(z)$ where $z$ is a root of $X^4+nX^3+2X^2-nX+1$. There the reflection removes the two conjugates $-1/z$ and $-1/w$, where the series diverge, and the remaining conjugate $w$ is a second convergent place, which enters through a norm. The result is a disjunction, proved separately for six values of $n$, with interval-arithmetic certificates. The smallest instance is the following.

**Theorem 1.3** (M1 at $n = 730$)**.** *Let $z > w$ be the two roots in $(0,1)$ of $X^4 + 730X^3 + 2X^2 - 730X + 1$. Then at least one of the following holds:*

$$\frac{\pi^2}{8} - \frac{1}{8}\log^2\bigl(366 + \sqrt{133955}\bigr) + \frac{1}{8}\log^2\bigl(364 + 11\sqrt{1095}\bigr) \notin \mathbb{Q},$$

$$\chi_2(z) - \chi_2(w) \notin \sqrt{1101}\,\mathbb{Q}.$$

The first number is $\chi_2(z) + \chi_2(w)$, by Landen's identity. Theorem 1.3 is not formalised.

1.5. **Organisation and notation.** An earlier version of this paper treated only $m = 2$. Section 2, Sections 3.1–3.3, Sections 5.1–5.3 and Section 6 are that version's arguments with $z = z_m$: they use only $\sigma(z) = -1/z$ and that $z$, $y$, $1-y$ are units at the primes concerned, and they hold verbatim for every $m$ except where the text says otherwise. New are Lemma S (Section 3.4), Lemma $10_\nu$ (Section 4), the shift lemma (Section 5.4), the hypothesis $\ell \nmid m$ in Lemma P′, and the margins (Section 7.2).

Section 2 sets up the functional, proves the pole expansion and the reflection, and states the criterion. Sections 3 and 4 bound the denominators, Section 5 bounds the size, Section 6 proves nonvanishing, and Section 7 assembles the proof and computes the margins. Section 8 discusses closed forms and compares Theorem 1.1 with known criteria. Section 9 treats two points, Section 10 describes the formalisation, and Section 11 lists the computations and the numerical evidence.

Throughout, $p$ and $\ell$ denote primes, $v_p$ is the $p$-adic valuation, $\lfloor\cdot\rfloor$ is the floor, $(x)_+ = \max(x, 0)$, and $[\,P\,]$ is 1 if $P$ holds and 0 otherwise. For a function $G$ on $\mathbb{Z}_{\geq 0}$, $\Delta G(s) = G(s+1) - G(s)$ is the forward difference. We write $\theta(x) = \sum_{p \leq x} \log p$ and $\psi(x) = \sum_{p^e \leq x} \log p$, and $s_2(b)$ for the binary digit sum of $b$, so $v_2(b!) = b - s_2(b)$. The letter $m$ always denotes the parameter of Theorem 1.1; poles are indexed by $k$.

## 2. The construction

2.1. **Data.** Fix an integer $m \geq 1$, put $d_m := m^2 + 1$ (not a square), let $\mathbb{K} = \mathbb{Q}(\sqrt{d_m})$, embedded in $\mathbb{R}$ with $\sqrt{d_m} > 0$, and let $\sigma$ be its nontrivial automorphism. With $z = z_m = \sqrt{d_m} - m$ and $y = z^2$ we have

$$z^2 + 2mz - 1 = 0, \qquad 1 - y = 2mz, \qquad \sigma(z) = -1/z.$$

So $z$ is a unit of $\mathcal{O}_\mathbb{K}$. At every odd prime $p \nmid m$ the elements $z$, $y$ and $1 - y$ are units; this includes the primes dividing $d_m$. At an odd prime $p \mid m$, $v_p(1 - y) = v_p(m)$. At every prime of $\mathbb{K}$ above 2, normalised by $v(2) = 1$, $v(1 - y) = \nu := 1 + v_2(m)$. The prime 2 is ramified in $\mathbb{K}$ for odd $m$, inert for $m \equiv 2 \bmod 4$ and split for $4 \mid m$: write $d_m = f^2 d_0$ with $d_0$ squarefree; $f$ is odd, so $d_0 \equiv d_m \bmod 8$, and $d_m \equiv 2, 5, 1 \bmod 8$ in the three cases. For $m = 2$: $\mathbb{K} = \mathbb{Q}(\sqrt{5})$, $z = \varphi^{-3}$, $1 - y = 4z$, $\nu = 2$.

Fix an even $h \geq 2$ and put $J := h/2$ and $K := h+1$. The poles are $x_k := 2(k-J)$ for $0 \leq k < K$, i.e. the even integers $\pm 2j$, $|j| \leq J$, and

$$D(t) := \prod_{k<K} (t - x_k),$$
$$D'(x_k) = \prod_{l \neq k} 2(k-l) = (-1)^{K-1-k} 2^{K-1} k!\,(K-1-k)!.$$

For a rational function $f$ regular at the odd positive integers and of polynomial growth put

$$\mathcal{M}(f) := \sum_{t \text{ odd} \geq 1} z^t f'(t),$$

an absolutely convergent series, and $\Lambda(g) := \mathcal{M}(g/D)$ for $g \in \mathbb{Q}[t]$. For an even integer $x$ define the partial sums

$$P(x) := \begin{cases} \displaystyle\sum_{n \text{ odd}, 0<n<|x|} \frac{z^n}{n^2}, & x < 0, \\ \displaystyle -\sum_{n \text{ odd}, -x<n<0} \frac{z^n}{n^2}, & x > 0, \\ 0, & x = 0. \end{cases}$$

Only odd $n$ with $|n| \leq |x| - 1 \leq h - 1$ occur.

## 2.2. **The pole expansion.**

**Lemma 2.1** (Lemma 1′, pole expansion)**.** *For every even integer $x$, $\mathcal{M}\big(1/(t-x)\big) = -z^x\big(T - P(x)\big)$. For a polynomial $F$,*

$$\mathcal{M}(F) = z \sum_{b \geq 0} \Delta^b G(0)\, \frac{y^b}{(1-y)^{b+1}}, \qquad G(s) := F'(2s+1), \tag{2.1}$$

*a finite sum. Consequently, writing $g/D = R_g + \sum_k g(x_k)/(D'(x_k)(t-x_k))$ with $R_g \in \mathbb{Q}[t]$,*

$$\Lambda(g) = A(g) + T\,B(g),$$
$$A(g) := \mathcal{M}(R_g) + \sum_k \frac{g(x_k)}{D'(x_k)} z^{x_k} P(x_k), \qquad B(g) := -\sum_k \frac{g(x_k)}{D'(x_k)} z^{x_k},$$

*with $A, B \colon \mathbb{Q}[t] \to \mathbb{K}$ linear.*

*Proof.* $\mathcal{M}(1/(t-x)) = -\sum_{t \text{ odd} \geq 1} z^t/(t-x)^2$. With $n = t - x$ (odd, $n \geq 1 - x$) and $z^t = z^x z^n$, the sum is $z^x \sum_{n \text{ odd} \geq 1-x} z^n/n^2 = z^x(T - P(x))$ by the definition of $P$. For a polynomial, substitute $t = 2s+1$, expand $G(s) = \sum_b \Delta^b G(0) \binom{s}{b}$, and use $\sum_{s \geq 0} \binom{s}{b} y^s = y^b/(1-y)^{b+1}$. All series converge absolutely since $0 < z < 1$. □

For an indeterminate $X$ put $\Lambda_X := A + X\,B$, $E_a := \Lambda_X(t^a) \in \mathbb{K}[X]$ and

$$\Delta_h(X) := \det\big(E_{i+j}\big)_{0\le i,j<h} \in \mathbb{K}[X], \qquad \deg \Delta_h \le h,$$
$$\Delta_h(T) = \det\big(\Lambda(t^{i+j})\big)_{i,j<h}.$$

We use the same notation for every finite family: if $a_0,\dots,a_{h-1}$ and $b_0,\dots,b_{h-1}$ have degrees $< h$ and coefficient matrices $C_a, C_b$, then $\det(\Lambda_X(a_i b_j)) = \det C_a \det C_b\, \Delta_h(X)$.

2.3. **The reflection.** Extend $A$ and $B$ to rational functions $f = R + \sum_k c_k/(t-x_k)$ ($R \in \mathbb{Q}[t]$, $c_k \in \mathbb{Q}$) by $A(f) := \mathcal{M}(R) + \sum_k c_k z^{x_k} P(x_k)$ and $B(f) := -\sum_k c_k z^{x_k}$, so that $\mathcal{M}(f) = A(f) + TB(f)$, and put $\tilde f(t) := f(-t)$.

**Lemma 2.2** (Lemma R, reflection)**.** *For every such $f$, $\sigma(A(f)) + \sigma(B(f))X = -\big(A(\tilde f) + B(\tilde f)X\big)$.*

*Proof.* We have $\tilde f = \tilde R + \sum_k (-c_k)/(t-(-x_k))$, and the pole set is symmetric. *The $X$-part.* Since $x_k$ is even, $\sigma(z^{x_k}) = (-1/z)^{x_k} = z^{-x_k}$, so $\sigma(B(f)) = -\sum_k c_k z^{-x_k} = -B(\tilde f)$. *The pole part of $A$.* For $x<0$, $\sigma(P(x)) = \sum_{0<n<|x|} (-1)^n z^{-n}/n^2 = -\sum_{-|x|<n'<0} z^{n'}/n'^2 = P(-x)$ (odd $n$, $n' = -n$), and likewise for $x>0$. Hence $\sigma\big(\sum_k c_k z^{x_k} P(x_k)\big) = \sum_k c_k z^{-x_k} P(-x_k)$, which is minus the pole part of $A(\tilde f)$. *The polynomial part.* For a polynomial $R$, $\mathcal{M}(R) = M_R(z)$ with $M_R(q) := \sum_{t\ \mathrm{odd}\ge 1} R'(t) q^t$, a rational function of $q$ with rational coefficients and poles only at $q = \pm 1$; so $\sigma(\mathcal{M}(R)) = M_R(-1/z)$. Since $R'$ is a polynomial, the reciprocity of rational generating functions gives $\sum_{t\ \mathrm{odd}\ge 1} R'(t) q^t = -\sum_{t\ \mathrm{odd}\le -1} R'(t) q^t$ as rational functions, the right side being expanded at $q=\infty$: for $R'(t) = t^a$ apply $(q\,d/dq)^a$ to $\sum_{t\ \mathrm{odd}\ge 1} q^t = q/(1-q^2) = -\sum_{t\ \mathrm{odd}\le -1} q^t$. At $q = -1/z$ the right side is $-\sum_{u\ \mathrm{odd}\ge 1} R'(-u)(-z)^u = \sum_{u\ \mathrm{odd}\ge 1} R'(-u) z^u = -M_{\tilde R}(z)$, because $\tilde R'(u) = -R'(-u)$. So $\sigma(\mathcal{M}(R)) = -\mathcal{M}(\tilde R)$. □

**Corollary 2.3** (Corollary R)**.** *$\sigma(E_a) = (-1)^a E_a$ coefficientwise, and $\Delta_h(X) \in \mathbb{Q}[X]$.*

*Proof.* For $f = t^a/D$ we have $D(-t) = (-1)^K D(t) = -D(t)$ ($K$ is odd), so $\tilde f = (-1)^{a+1} f$, and Lemma 2.2 gives $\sigma(E_a) = (-1)^a E_a$. Thus $E_a \in \mathbb{Q}[X]$ for even $a$ and $E_a \in \sqrt{d_m}\,\mathbb{Q}[X]$ for odd $a$. The matrix $\tilde E_{ij} := E_{i+j} \sqrt{d_m}^{\,[i\ \mathrm{odd}]-[j\ \mathrm{odd}]}$ has entries in $\mathbb{Q}[X]$, and $\tilde E = \mathrm{diag}(\sqrt{d_m}^{\,[i\ \mathrm{odd}]})\,(E_{i+j})\,\mathrm{diag}(\sqrt{d_m}^{\,-[j\ \mathrm{odd}]})$, so $\det \tilde E = \Delta_h$. □

Lemma 2.2 and Corollary 2.3 use only $\sigma(z) = -1/z$; whether $d_m$ is squarefree plays no role.

2.4. **Valuations and the criterion.** For a prime $p$ and a prime $\mathfrak{p} \mid p$ of $\mathbb{K}$ let $v_\mathfrak{p}$ be normalised by $v_\mathfrak{p}(p) = 1$ (so $v_\mathfrak{p}$ takes values in $\frac{1}{2}\mathbb{Z}$ at ramified

primes), extended to $\mathbb{K}[X]$ as the Gauss valuation (the minimum over the coefficients). On $\mathbb{Q}[X]$ every $v_\mathfrak{p}$ restricts to $v_p$. For $\Delta_h \not\equiv 0$ put

$$n_p := -v_p(\Delta_h), \qquad F_h := \Delta_h \prod_p p^{n_p} \in \mathbb{Z}[X] \text{ (primitive)},$$

$$\log|F_h(T)| = \log|\Delta_h(T)| + \sum_p n_p \log p.$$

Upper bounds for $n_p$ may be computed from the $\mathbb{K}$-valued entries at any one $\mathfrak{p} \mid p$, since $\Delta_h \in \mathbb{Q}[X]$.

**Proposition 2.4** (Lemma E)**.** *Let $\xi \in \mathbb{R}$ and let $\mathcal{H}$ be an infinite set of positive integers. Suppose that for $h \in \mathcal{H}$ there are $F_h \in \mathbb{Z}[X]$ with $\deg F_h \leq h$ and $\limsup_{h\in\mathcal{H}} h^{-2}\log|F_h(\xi)| < 0$, and that $F_h(a/b) \neq 0$ for all $h \in \mathcal{H}$, where $a \in \mathbb{Z}$, $b \geq 1$. Then $\xi \neq a/b$.*

*Proof.* If $\xi = a/b$, then $b^h F_h(a/b)$ is a nonzero integer, so $h\log b + \log|F_h(\xi)| \geq 0$ for $h \in \mathcal{H}$. Dividing by $h^2$ contradicts the hypothesis. □

This is the criterion of [15, Prop. 3.1], with the nonvanishing set allowed to depend on $b$.

## 3. Denominators I: the odd primes

In Sections 3.1–3.3, $p$ is an odd prime with $p \nmid m$, and $v = v_\mathfrak{p}$ for a prime $\mathfrak{p} \mid p$ of $\mathbb{K}$. Odd primes dividing $m$ are treated in Section 3.4. Since $D$ is monic with integer coefficients, $R_g \in \mathbb{Z}[t]$ for $g \in \mathbb{Z}[t]$, and by (2.1), with $y$ and $1-y = 2mz$ units at $\mathfrak{p}$,

$$v\big(\mathcal{M}(F)\big) \geq \min_{t \text{ odd}\geq 1} v_p\big(F'(t)\big) \qquad (F \in \mathbb{Z}_{(p)}[t]), \tag{3.1}$$

because $\Delta^b G(0)$ is an integer combination of values $F'(t)$ at odd $t$. For odd $t$ we have $t - x_k = (t+2J) - 2k$, so for a level $p^e$ the indices $k$ with $p^e \mid t - x_k$ form one residue class $k_0(t)$ modulo $p^e$ among the $K$ consecutive integers $0, \dots, K-1$. Put

$$\mathrm{lev}^*(p) := \#\{e \geq 1 : p^e \leq h\}.$$

3.1. **The atom expansion and the class count.** Let $V_{k,i} := x_k^i$ ($k < K$, $i < h$), $\mathcal{P}_{ij} := \mathcal{M}(R_{t^{i+j}})$ and $\gamma_k(X) := -z^{x_k}(X - P(x_k))/D'(x_k)$. By Lemma 2.1, $(E_{i+j}) = \mathcal{P} + V^{\mathsf{T}} \operatorname{diag}(\gamma) V$.

**Lemma 3.1** (Lemma 7′, atom expansion)**.**

$$\Delta_h(X) = \sum_{|I|=|I'|=|S|} \pm \det \mathcal{P}_{I^c I'^c} \det V_{S,I} \det V_{S,I'} \prod_{k\in S} \gamma_k(X),$$

*over $I, I' \subseteq \{0, \dots, h-1\}$ and $S \subseteq \{0, \dots, K-1\}$. Moreover $\det V_{S,I} \in \mathrm{Vand}(x_S)\mathbb{Z}$, and $\prod_{k\in S}|D'(x_k)| = \mathrm{Vand}(x_S)^2 \operatorname{cross}(S)$ with $\operatorname{cross}(S) := \prod_{k\in S, l\notin S}|x_k - x_l|$.*

*Proof.* Expand $\det(\mathcal{P}+M)$ by the complementary minors of $\mathcal{P}$ and $M = V^{\mathsf{T}}\operatorname{diag}(\gamma)V$, and each minor $\det M_{II'}$ by the Cauchy–Binet formula. A generalised Vandermonde determinant is $\operatorname{Vand}(x_S)$ times a Schur polynomial with integer coefficients. The last identity splits $\prod_{l\neq k}|x_k-x_l|$ into $l\in S$ and $l\notin S$. □

Lemma 3.1 is algebra and holds at every prime. For an odd level $d$ put $w_d(k) := 2[\,|x_k| > d\,]$, let $C_c := \{k < K : k \equiv c \bmod d\}$, $n_c := |C_c|$, and

$$A_K(d) := \sum_{c \bmod d} \max_{S_c \subseteq C_c} \Big[ |S_c|(n_c - |S_c|) + \sum_{k\in S_c} w_d(k) \Big].$$

**Lemma 3.2** (Lemma 8′, per-level bound)**.** *For every odd prime $p \nmid m$, $n_p \le \sum_{e\ge 1,\, p^e<K} A_K(p^e)$.*

*Proof.* $\mathcal{P}$ is $\mathfrak{p}$-integral by (3.1). In the term of Lemma 3.1 indexed by $(I, I', S)$, the factors $\det V_{S,I}\det V_{S,I'}$ cancel $\operatorname{Vand}(x_S)^2$, so by Lemma 3.1 the term has valuation at least $v(\det \mathcal{P}_{I^cI'^c}) - v_p(\operatorname{cross}(S)) + \sum_{k\in S} v(X - P(x_k))$, using that $z^{x_k}$ is a unit. Here $v_p(\operatorname{cross}(S)) = \sum_{e\ge1} \#\{(k,l) : k\in S,\ l\notin S,\ p^e \mid k-l\}$ since $x_k - x_l = 2(k-l)$, and $v(X-P(x_k)) \ge -2\max_{n \text{ odd}, |n|<|x_k|} v_p(n) \ge -2\sum_{e\ge1}[\,p^e < |x_k|\,]$. At level $d = p^e$ the pairs counted are those inside residue classes, $\sum_c |S\cap C_c|(n_c - |S\cap C_c|)$, and $[\,p^e < |x_k|\,] = \frac{1}{2}w_d(k)$. Bounding the maximum over $S$ level by level and class by class gives the claim. Levels $p^e \ge K$ contribute nothing: the classes are singletons and $|x_k| \le h < p^e$. □

**Lemma 3.3** (Lemma 8A′, class count)**.** *For odd $d$ with $K/3 \le d < K$ we have $A_K(d) = 3(K-d)$, and $A_K(d) = 0$ for $d \ge K$.*

*Proof.* For $d \ge K$ all classes are singletons and all weights vanish. Let $K/3 \le d < K$. Then $w_d(k) = 0$ if and only if $|k - J| \le (d-1)/2$, a block of $d$ consecutive indices inside $[0, K)$; so every class contains exactly one atom of weight 0, and the others have weight 2. A class with $n$ atoms contributes $V(n) := \max_t[t(n-t) + 2\min(t, n-1)]$ (take the weight-0 atom last), and $V(1) = 0$, $V(2) = 3$, $V(3) = 6$. Since $3d \ge K$, classes have at most three atoms. If $K/3 \le d \le K/2$ there are $K - 2d$ classes of size 3 and $3d - K$ of size 2, with total $6(K-2d) + 3(3d-K) = 3(K-d)$. If $K/2 < d < K$ there are $K - d$ pairs and $2d - K$ singletons, with total $3(K-d)$. □

3.2. **The plateau: a bordered Lagrange Gram identity.** In [15] the number of poles equals the size $h$ of the determinant, and the Lagrange basis has the right dimension. Here $K = h+1$, and the $K$ Lagrange polynomials $f_i := D/(t-x_i)$ have degree $h$: they span the polynomials of degree $\le h$, one dimension too many. The adjugate of the $(h+1)$-Gram matrix repairs this.

**Lemma 3.4** (Lemma B, bordered identity)**.** *Let* $G^f(X) := (\Lambda_X(f_i f_j))_{i,j<K}$. *Then*

$$\Delta_h(X) = (-1)^{K(K-1)/2}\,\frac{\mathbf{1}^{\mathsf{T}}\operatorname{adj}\big(G^f(X)\big)\mathbf{1}}{\prod_{i<K} D'(x_i)}.$$

*Proof.* Let $H_{h+1} := (E_{a+b})_{a,b\leq h}$. Then $\Delta_h = \operatorname{adj}(H_{h+1})_{hh}$. Let $C$ be the coefficient matrix of the $f_i$ (row $i$ lists the coefficients of $t^0,\dots,t^h$ in $f_i$). Then $G^f = CH_{h+1}C^{\mathsf{T}}$, so $\operatorname{adj}(G^f) = (\det C)^2 C^{-\mathsf{T}}\operatorname{adj}(H_{h+1})C^{-1}$ and $\operatorname{adj}(H_{h+1}) = (\det C)^{-2}C^{\mathsf{T}}\operatorname{adj}(G^f)C$. Column $h$ of $C$ is $\mathbf{1}$, since every $f_i$ is monic of degree $h$; hence $\Delta_h = (\det C)^{-2}\mathbf{1}^{\mathsf{T}}\operatorname{adj}(G^f)\mathbf{1}$. Finally $(f_i(x_k))_{i,k} = \operatorname{diag}(D'(x_i)) = CW$ with $W = (x_k^a)_{a,k}$, and $\prod_i D'(x_i) = (-1)^{K(K-1)/2}(\det W)^2$, so $(\det C)^2 = (-1)^{K(K-1)/2}\prod_i D'(x_i)$. □

**Lemma 3.5** (Lemma 9′, plateau bound)**.** *For every odd prime* $p \nmid m$, $n_p \leq 2h\,\mathrm{lev}^*(p) + v_p(h!)$.

*Proof.* Put $N_e(i) := \#\{k < K : k \neq i,\ p^e \mid k-i\}$ and $N(i) := \sum_e N_e(i) = v_p(D'(x_i))$. Note that $N_e(i) \geq 1$ needs $p^e \leq K-1 = h$. We first prove the *row claim*

$$v\big(G^f_{ij}\big) \geq N(i) - 2\,\mathrm{lev}^*(p) \qquad (i,j<K). \tag{3.2}$$

Fix an odd $t$ and a level $d = p^e$, and let $k_0 = k_0(t)$. The class of $i$ modulo $d$ has $N_e(i)+1$ elements in $[0,K)$, and classes modulo $d$ in $[0,K)$ differ in size by at most one, so the class $k_0$ has at least $N_e(i)$ elements.

*Off-diagonal entries* ($i \neq j$). Here $f_i f_j/D = F := \prod_{k\neq i,j}(t-x_k)$, so $G^f_{ij} = \mathcal{M}(F)$. Each term of $F'(t)$ omits one factor. At level $d$ the factors divisible by $d$ are those with $k \equiv k_0$, $k \neq i,j$; after the omission at least $N_e(i)-2$ remain, whether $k_0 \equiv i$ or not. Summing over levels, every term has $v_p \geq \sum_e (N_e(i) - 2[\,N_e(i) \geq 1\,]) \geq N(i) - 2\,\mathrm{lev}^*(p)$, and (3.1) gives (3.2).

*Diagonal entries.* Put $g := \prod_{k\neq i}(t-x_k)$. Then $f_i^2/D = g/(t-x_i) = Q + g(x_i)/(t-x_i)$ with $Q := (g(t)-g(x_i))/(t-x_i)$ and $g(x_i) = D'(x_i)$. By Lemma 2.1 the pole part contributes $-D'(x_i)z^{x_i}(X - P(x_i))$, of valuation at least $N(i) - 2\#\{e : p^e < |x_i|\} \geq N(i) - 2\,\mathrm{lev}^*(p)$, since $|x_i| \leq h$. For $Q$, list the roots $x_k$, $k \neq i$, of $g$ as $\alpha_1,\dots,\alpha_h$ in order of decreasing $v_p(k-i)$; telescoping gives

$$Q(t) = \sum_{r=1}^{h}\prod_{s<r}(x_i-\alpha_s)\prod_{s>r}(t-\alpha_s),$$

and for every level $e$ the $\alpha_s = x_k$ with $p^e \mid k-i$ form an initial segment of length $N_e(i)$ (note $x_i - x_k = 2(i-k)$). Fix a term $r$ of $Q'(t)$, which omits one more factor of the second product, and a level $d$. If $r > N_e(i)$, the first product has $N_e(i)$ factors divisible by $d$. If $r \leq N_e(i)$ and $k_0 \equiv i$, it has $r-1$, and the remaining $N_e(i)-r$ class members sit at positions $s > r$, at most one of them omitted: in total at least $N_e(i)-2$. If $r \leq N_e(i)$ and $k_0 \not\equiv i$, the whole class $k_0$ (at least $N_e(i)$ members, all with $v_p(k-i) < e$)

sits at positions $s > N_e(i) \geq r$: in total at least $N_e(i) - 1$. So (3.2) holds for $Q$ too, by (3.1).

*The minors.* $\mathbf{1}^{\mathsf{T}} \operatorname{adj}(G^f)\mathbf{1} = \sum_{a,b} \pm M_{ba}$, where $M_{ba}$ is the minor with row $b$ and column $a$ deleted. By (3.2) and the Leibniz expansion, $v(M_{ba}) \geq \sum_{i \neq b}(N(i) - 2\operatorname{lev}^*(p)) \geq \sum_i N(i) - 2h\operatorname{lev}^*(p) - \max_b N(b)$. Lemma 3.4 and $v_p(\prod_i D'(x_i)) = \sum_i N(i)$ give $v(\Delta_h) \geq -2h\operatorname{lev}^*(p) - \max_b N(b)$. Finally $N(b) = v_p(b!\,(h-b)!) \leq v_p(h!)$. □

*Remark* 3.6. At $m = 2$, on the exact profiles computed for $h = 16, \ldots, 256$, the bound of Lemma 3.5 is never violated, and it is attained at most single-level plateau primes; for example at $h = 128$ the primes $13, 17, 19, 23, 31, 37, 41, 43$ have $n_p = 256 + v_p(128!)$ exactly. The row claim (3.2) was checked entrywise at $h = 16, 24, 32, 40$ and $p \leq 13$, with minimum slack 0. For other $m$, Lemmas 3.2 and 3.5 were checked at every odd $p \nmid m$ on exact profiles for $m = 3, 4, 5, 6, 9, 15$ and $h \leq 48$, and in 47 further pairs $(m, h)$ by an independent implementation; there was no violation.

### 3.3. The sum over the odd primes not dividing $m$.

**Theorem 3.7** (Theorem A′)**.** *For every $\varepsilon > 0$ and all large even $h$, $\sum_{p \text{ odd}, p \nmid m} n_p \log p \leq (\frac{4}{3} + \varepsilon)h^2$.*

*Proof.* For $p < K/3$ use Lemma 3.5:

$$\sum_{p < K/3} \big(2h\operatorname{lev}^*(p) + v_p(h!)\big)\log p \leq 2h\big(\theta(K/3) + \psi(h) - \theta(h)\big) + \log h!$$
$$= \tfrac{2}{3}h^2 + o(h^2)$$

by the prime number theorem. For $K/3 \leq p < K$ we have $p^2 \geq K^2/9 \geq K$ for $h \geq 8$, so only the level $e = 1$ occurs, and Lemmas 3.2 and 3.3 give $n_p \leq 3(K - p)$. By the prime number theorem and partial summation, $\sum_{K/3 \leq p < K} 3(K-p)\log p = 3h^2 \int_{1/3}^1 (1-x)\,dx + o(h^2) = \frac{2}{3}h^2 + o(h^2)$. For $p \geq K$, $n_p \leq 0$ by Lemma 3.2. □

The finite sums of these bounds over all odd primes, divided by $h^2$, are 1.3380, 1.3350 and 1.3337 at $h = 1442$, 20836 and 875032. The primes dividing $d_m$ (for $m = 2$, the prime 5) need no separate treatment: $z$ and $1 - y = 2mz$ are units there, so every step above holds with $v(p) = 1$.

### 3.4. Odd primes dividing $m$: Lemma S.

At an odd prime $p \mid m$ the matrix $\mathcal{P}$ is no longer $\mathfrak{p}$-integral, since $v_p(1 - y) = v_p(m) > 0$. The bound of Lemma 3.2 survives with an extra term.

**Lemma 3.8** (Lemma S)**.** *Let $p$ be an odd prime with $e := v_p(m) \geq 1$. Then for every even $h$,*

$$n_p \leq \sum_{l \geq 1,\, p^l < K} A_K(p^l) + e\,\frac{(h-2)^2}{4}.$$

*Consequently* $\limsup_{h\to\infty} n_p/h^2 \leq \frac{e}{4} + \frac{1}{4(p-1)}$, *i.e. the prime* $p$ *costs at most* $c_p(e) = \big(\frac{e}{4} + \frac{1}{4(p-1)}\big)\log p$ *per* $h^2$.

*Proof.* Lemma 3.1 still holds. In the proof of Lemma 3.2 every factor other than $\det \mathcal{P}_{I^c I'^c}$ is treated as before: $V$ is integral, $z^{x_k}$ is a unit, $P(x)$ has denominators $n^2$ with $n$ odd, and the class count does not involve $\mathcal{P}$. This gives $v(\text{term}) \geq v(\det \mathcal{P}_{I^c I'^c}) - \sum_l A_K(p^l)$. For the polynomial part, $\mathcal{P}_{ij} = \mathcal{M}(R_{t^{i+j}})$, where $R_{t^{i+j}} \in \mathbb{Z}[t]$ is the quotient of $t^{i+j}$ by $D$ (monic of degree $K$). So $\mathcal{P}_{ij} = 0$ for $i+j \leq K$, since then $R$ is constant and $\mathcal{M}$ of a constant is 0. For $r := i+j-K \geq 1$, the Mahler form (2.1) with $G(s) = R'(2s+1) \in \mathbb{Z}[s]$ of degree $r-1$, and $\Delta^b G(0) \in b!\,\mathbb{Z}$, gives

$$v(\mathcal{P}_{ij}) \geq \min_{0\leq b\leq r-1} \big(v_p(b!) - e(b+1)\big) \geq -e\,r.$$

So $v(\mathcal{P}_{ij}) \geq -e(i+j-K)_+$. For any bijection $\tau\colon I^c \to I'^c$, $(i+\tau(i)-K)_+ \leq (i-K/2)_+ + (\tau(i)-K/2)_+$, hence

$$v(\det \mathcal{P}_{I^c I'^c}) \geq -2e \sum_{i<h} (i-K/2)_+ = -e(h/2-1)^2 = -e\,\frac{(h-2)^2}{4}$$

($h$ even, $K = h+1$). Adding the two bounds gives the first claim. For the limit, a class with $n_c \leq K/d+1$ atoms contributes at most $\max_s[s(n_c-s)+2s] \leq (K/d+3)^2/4$ to $A_K(d)$, so $A_K(d) \leq d(K/d+3)^2/4 = K^2/(4d)+O(K)$, and $\sum_{l\geq 1} p^{-l} = 1/(p-1)$; there are $O(\log K)$ levels. □

*Checks.* On exact profiles there was no violation. For $p=3$, $e=1$ (at $m=3$, 6 and 15, where $n_3$ was the same): $n_3 = 101,202,355,752$ at $h = 16, 24, 32, 48$, against the bounds $117, 249, 441, 971$. For $m=9$ ($e=2$): $n_3 = 150,579$ at $h = 16, 32$, against $166,666$. At $m=6$, $h=48$, the generic part of $n_3$ (the value 279 at $m=2$) plus $e(h-2)^2/4 = 529$ gives $808 \geq 752$, so the term $e(h-2)^2/4$ is close to sharp, while the plateau term of Lemma 3.2 (442) overshoots the generic 279. These data suggest that the true cost of a special prime is about $\frac{e}{4}\log p$ per $h^2$, and that the extra $\frac{1}{4(p-1)}\log p$ comes from using Lemma 3.2 instead of Lemma 3.5 at $p$; we have not proved this (see Remark 7.2).

## 4. Denominators II: the prime 2

Let $\mathfrak{P}$ be a prime of $\mathbb{K}$ above 2 and $v := v_{\mathfrak{P}}$, normalised by $v(2) = 1$. Then $v(z) = 0$ and $v(1-y) = \nu = 1 + v_2(m)$ (Section 2.1). For $m = 2$, 2 is inert and $\nu = 2$.

*Rescaling.* Put $\tau := t/2$. The atoms become $\tau = s + \frac{1}{2}$ ($s \geq 0$), the poles $\xi_k := k - J \in \mathbb{Z}$, and $D(t) = 2^K \tilde{D}(\tau)$ with $\tilde{D}(\tau) := \prod_{k<K}(\tau - \xi_k)$. With $\tilde{\mathcal{M}}(F) := \sum_{s\geq 0} z^{2s+1} F'(s+\frac{1}{2})$ and $\tilde{\Lambda}(g) := \tilde{\mathcal{M}}(g/\tilde{D})$ we get $\Lambda(t^a) = 2^{a-K-1}\tilde{\Lambda}(\tau^a)$, hence $\Delta_h = 2^{-3h}\tilde{\Delta}_h$ with $\tilde{\Delta}_h := \det(\tilde{\Lambda}_X(\tau^{i+j}))$, since $\sum_{i<h} 2i - h(K+1) = -3h$.

*Anchored basis.* Put $a_i := \prod_{l<i}(\tau-\xi_l)$ and $b_j := \prod_{l<j}(\tau-\xi_{K-1-l})$. Both families are monic of degrees $i$ and $j$, so $\det(G_{ij}) = \tilde{\Delta}_h$ for $G_{ij} := \tilde{\Lambda}_X(a_ib_j)$.

4.1. **The entries.**

**Lemma 4.1** (Lemma $10_\nu$)**.** *Let $i, j < h$.*

(L) *If $\pi := i + j - K \geq 1$, then $v(G_{ij}) \geq 1 - \pi - \nu + \min_{0\leq b\leq \pi-1}\big((2-\nu)b - s_2(b)\big)$. If $\pi = 0$, then $G_{ij} = 0$.*

(U) *If $N_0 := K - 1 - i - j \geq 1$, then*

$$v(G_{ij}) \geq \min\Big\{2 + (\nu-1)N_0 + s_2(N_0),\\ 2 - N_0 + s_2(N_0) + \min_{0\leq r\leq N_0-1}\big(2r - s_2(r) + \nu(N_0-1-r)\big)\Big\}.$$

*If $N_0 = 0$, then $v(G_{ij}) = 2$.*

*Proof.* (L) The products $a_i$ and $b_j$ cover the index ranges $[0, i-1]$ and $[K-j, K-1]$, which overlap in $\pi$ indices, so $a_ib_j/\tilde{D} = F := \prod_{l=K-j}^{i-1}(\tau-\xi_l) = 2^{-\pi}H(2\tau)$ with $H(U) := \prod_l(U-2\xi_l) \in \mathbb{Z}[U]$. So $F'(s+\frac{1}{2}) = 2^{1-\pi}H'(2s+1)$, and by (2.1) (with $s$ in place of $(t-1)/2$), $\tilde{\mathcal{M}}(F) = z\sum_{b<\pi}\Delta^bG(0)y^b/(1-y)^{b+1}$, $G(s) = 2^{1-\pi}H'(2s+1)$. The Newton coefficients of $s \mapsto (2s+1)^a$ are $\Delta^b[(2s+1)^a](0) = \sum_{j'\geq b}\binom{a}{j'}2^{j'}\,b!\,S(j',b)$, of valuation $\geq b + v_2(b!) = 2b - s_2(b)$, while $y^b/(1-y)^{b+1}$ has valuation $-\nu(b+1)$. Hence $v(G_{ij}) \geq 1 - \pi + \min_{b<\pi}\big(2b - s_2(b) - \nu(b+1)\big)$, which is the claim. There is no $X$-part. If $\pi = 0$ then $F = 1$ and $\tilde{\mathcal{M}}(1) = 0$.

(U) Now the ranges leave a gap of $N_0 + 1$ indices, and with $\zeta_l := \xi_i + l$,

$$\frac{a_ib_j}{\tilde{D}} = \prod_{l=0}^{N_0}\frac{1}{\tau-\zeta_l} = \frac{1}{N_0!}\sum_{l=0}^{N_0}(-1)^{N_0-l}\binom{N_0}{l}\frac{1}{\tau-\zeta_l}.$$

For an integer $\zeta$, $\tilde{\mathcal{M}}(1/(\tau-\zeta)) = -4\sum_s z^{2s+1}/(2s+1-2\zeta)^2 = -4z^{2\zeta}(X - S(\zeta))$ (at $X = T$), where $S(\zeta+1) - S(\zeta) = -z^{-1-2\zeta}/(2\zeta+1)^2$; so $S(\zeta)$ has only odd denominators and $v(S(\zeta)) \geq 0$. With $z^{2\zeta_l} = z^{2\xi_i}y^l$ and $S(\zeta_l) = S(\xi_i) - \sum_{r<l}z^{-1-2\xi_i-2r}g(r)$, $g(r) := (s_0+2r)^{-2}$, $s_0 := 2\xi_i + 1$,

$$G_{ij} = -\frac{4}{N_0!}\Big[z^{2\xi_i}(y-1)^{N_0}\big(X - S(\xi_i)\big) + z^{-1}\sum_l(-1)^{N_0-l}\binom{N_0}{l}\sum_{r<l}y^{l-r}g(r)\Big].$$

The first part of the bracket has valuation $\nu N_0$. In the second, write $y^{l-r} = y\sum_{e'}\binom{l-r-1}{e'}\eta^{e'}$ with $\eta := y - 1$ ($v(\eta) = \nu$) and $g(r) = \sum_a\gamma_a\binom{r}{a}$; expanding $g(r) = s_0^{-2}\sum_c(-1)^c(c+1)(2/s_0)^cr^c$ (convergent in $\mathbb{Z}_2$) and using $\Delta^ar^c(0) = a!\,S(c,a)$ gives $v_2(\gamma_a) \geq a + v_2(a!) = 2a - s_2(a)$. The identities $\sum_{r<l}\binom{r}{a}\binom{l-1-r}{e'} = \binom{l}{a+e'+1}$ and $\sum_l(-1)^{N_0-l}\binom{N_0}{l}\binom{l}{M} = [\,M = N_0\,]$ collapse the second part to $z^{-1}y\sum_{a+e'=N_0-1}\gamma_a\eta^{e'}$, of valuation $\geq \min_{0\leq a\leq N_0-1}(2a - s_2(a) + \nu(N_0-1-a))$. With $v_2(N_0!) = N_0 - s_2(N_0)$ and $v(4) = 2$ this gives (U). If $N_0 = 0$ then $G_{ij} = -4z^{2\xi_i}(X - S(\xi_i))$, of valuation exactly 2. □

Nothing in this proof uses a particular value of $\nu$; for $\nu = 2$ it is the computation of the earlier version.

4.2. **The content.**

**Proposition 4.2.** *For every even $h \geq 2$:*

$$n_2 \leq \begin{cases} (\nu-1)\dfrac{(h-2)^2}{4} - \dfrac{h^2}{4} + h\lfloor \log_2 h \rfloor + 5h - 1, & \nu \geq 2, \\ \dfrac{(h-2)^2}{4} + 3h, & \nu = 1. \end{cases}$$

*Hence* $\limsup_h n_2/h^2 \leq |\nu-2|/4$, *and the prime* 2 *costs at most* $c_2(\nu) = |\nu-2|\frac{\log 2}{4}$ *per* $h^2$.

*Proof.* $\nu \geq 2$. In (L), $(2-\nu)b \geq (2-\nu)(\pi-1)$ and $s_2(b) \leq \lfloor \log_2 \pi \rfloor + 1$, so $v(G_{ij}) \geq -(\nu-1)\pi - 2 - \lfloor \log_2 \pi \rfloor$. In (U), $2r - s_2(r) + \nu(N_0 - 1 - r) \geq 2(N_0 - 1) - s_2(r)$, so $v(G_{ij}) \geq N_0 - 1 + s_2(N_0) - \lfloor \log_2 N_0 \rfloor \geq N_0 - \lfloor \log_2 N_0 \rfloor$; the first term of the minimum in (U) is larger. Hence $v(G_{ij}) \geq \hat{g}(i+j) - \lfloor \log_2 h \rfloor - 2$ for all $i, j$, with the piecewise linear function

$$\hat{g}(s) := K - 1 - s \quad (s \leq K), \qquad \hat{g}(s) := -1 - (\nu-1)(s-K) \quad (s \geq K),$$

which is concave since its slopes $-1$ and $-(\nu-1)$ decrease (at $s = K$ the entry vanishes, and at $s = K-1$ its valuation is $2 \geq 0$). For a permutation $\sigma$ of $\{0, \ldots, h-1\}$ the vector $(i+\sigma(i))_i$ is majorised by $(2i)_i$: the $k$ largest entries of $i + \sigma(i)$ sum to at most twice the $k$ largest $i$, and the totals are equal. By Karamata's inequality, $\sum_i \hat{g}(i+\sigma(i)) \geq \sum_i \hat{g}(2i) = \frac{h^2}{4} + 1 - (\nu-1)\frac{(h-2)^2}{4}$. By the Leibniz expansion, $v(\tilde{\Delta}_h) \geq \frac{h^2}{4} + 1 - (\nu-1)\frac{(h-2)^2}{4} - h(\lfloor \log_2 h \rfloor + 2)$, and $n_2 = -v(\Delta_h) = 3h - v(\tilde{\Delta}_h)$.

$\nu = 1$. (L) gives $v(G_{ij}) \geq -\pi$, since $\min_b(b - s_2(b)) = 0$. In (U) the first term is $2 + s_2(N_0)$ and the second is $1 + s_2(N_0) + \min_r(r - s_2(r)) = 1 + s_2(N_0) \geq 2$. So $v(G_{ij}) \geq \hat{g}_1(i+j)$ with the concave $\hat{g}_1(s) := -(s-K)_+$, and $\sum_i \hat{g}_1(2i) = -(1 + 3 + \cdots + (h-3)) = -(h/2-1)^2$. Conclude as before. □

At $\nu = 3$ (for example $m = 4$) the bound reads $n_2 \leq h^2/4 + h\lfloor \log_2 h \rfloor + 3h + 1$. On exact profiles at $m = 4$, $n_2 = 23, 94, 201, 349, 523, 737, 1277$ at $h = 8, 16, 24, 32, 40, 48, 64$, against the bounds $65, 177, 313, 513, 721, 961, 1601$; the share $n_2 \log 2/h^2$ is 0.249 at $h = 8$ and 0.216 at $h = 64$, and 0.2046, 0.2002 at $h = 96$, 128, falling towards the proved limit 0.1733. At $m = 8$ ($\nu = 4$), $n_2 = 2205$ at $h = 64$ against the bound 2562. Every entry bound (L), (U) was checked on the exact Gram matrices for $m = 1, 3, 4, 6, 8, 12, 16$ and $h \leq 64$ (both embeddings where 2 splits), with no violation and with equality attained. In [15] the corresponding quantity is $v_2(1 - q^{-2}) = 3$, which costs $\frac{1}{4}\log 2$ per $h^2$, in agreement with $c_2(3)$.

4.3. **The case $\nu = 2$.** At $\nu = 2$ Proposition 4.2 gives $n_2 \le h\lfloor \log_2 h\rfloor + 4h$. The direct assignment gives a slightly better bound; for $m = 2$ this is the bound used in the earlier version.

**Lemma 4.3** (Lemma 10′)**.** *Let $\nu = 2$ (that is, $m \equiv 2 \bmod 4$) and $i, j < h$.*

(L) *If $\pi \ge 1$, then $v(G_{ij}) \ge -\pi - \lfloor \log_2 \pi\rfloor - 2$. If $\pi = 0$, then $G_{ij} = 0$.*
(U) *If $N_0 \ge 1$, then $v(G_{ij}) \ge N_0 - \lfloor \log_2 N_0\rfloor$. If $N_0 = 0$, then $v(G_{ij}) = 2$.*

*Consequently $n_2 \le h\lfloor \log_2 h\rfloor + 3h$.*

*Proof.* (L) and (U) are Lemma 4.1 at $\nu = 2$: in (L), $1 - \pi - 2 + \min_{b<\pi}(-s_2(b)) \ge -\pi - \lfloor \log_2 \pi\rfloor - 2$; in (U), the second term is $N_0 - 1 + s_2(N_0) - \lfloor \log_2 N_0\rfloor$ and the first, $N_0 + 2 + s_2(N_0)$, is larger. *Assignment.* In all cases $v(G_{ij}) \ge K - i - j - \lfloor \log_2 h\rfloor - 2$, since $\pi, N_0 \le h$. By the Leibniz expansion, $v(\tilde\Delta_h) \ge \sum_{i<h}(K - i - \sigma(i)) - h(\lfloor \log_2 h\rfloor + 2) = -h\lfloor \log_2 h\rfloor$ for every permutation $\sigma$, and $n_2 = 3h - v(\tilde\Delta_h)$. □

The mechanism is $\nu = 2$: both triangles (L) and (U) then have slope 1 in $i + j$, and the permutation sum of the bound is constant. Section 9 shows what happens when $\nu = 1$ in a biquadratic setting. At $m = 2$ the exact values $n_2 = 54, 96, 142, 240, 350, 576, 830$ at $h = 16, 24, 32, 48, 64, 96, 128$ lie below the bound of Lemma 4.3 $(112, 168, 256, 384, 576, 864, 1280)$, and $n_2/(h \log_2 h)$ rises from 0.84 to 0.94 for $h \le 256$.

## 5. The size of $\Delta_h(T)$: Theorem B′

5.1. **The equilibrium problem and weak duality.** For $0 < z < 1$ put

$$\Psi(x) := \tfrac{1}{2}\int_{x-1}^{x+1} \log|s|\, ds, \qquad v_z(x) := x \log z - \Psi(x).$$

$\Psi$ is even, increasing on $[0, \infty)$, and $\Psi(0) = -1$. The function $v_z$ is continuous, and $v_z'(x) = \log z - \frac{1}{2}\log|(x+1)/(x-1)|$ for $x \ne \pm 1$. For $x > 0$, $x \ne 1$, the logarithm is positive, so $v_z' < \log z < 0$; for $-1 < x \le 0$, $v_z'(x) = \log z + \operatorname{artanh}|x| < 0$ if and only if $|x| < \tanh(\log(1/z))$. For $0 < z \le \sqrt{5} - 2$ we have $\tanh(\log(1/z)) \ge \tanh(\log(2 + \sqrt{5})) = 0.8945\ldots$. Hence

$$v_z \text{ is strictly decreasing on } (-0.894, \infty) \supset [-1/h, \infty) \tag{5.1}$$

for every $h \ge 2$ and $0 < z \le \sqrt{5} - 2$. This covers every $z_m$ with $m \ge 2$, since $z_m$ decreases in $m$. For a finite positive measure $\rho$ with compact support put $U^\rho(x) := \int \log|x - s|\, d\rho(s)$ and $E(\rho) := \iint \log|x - s|\, d\rho(x)\, d\rho(s)$.

Call $\mu$ *admissible* if it is a density with $0 \le \mu \le 1$, mass 1 and $\operatorname{supp}\mu \subset [0, 2]$. For admissible $\mu$ and $\lambda \in \mathbb{R}$ put

$$F_\mu := 2U^\mu + v_z, \qquad \Phi_z(\mu, \lambda) := -E(\mu) + \lambda + \int_0^\infty (F_\mu - \lambda)_+\, dx.$$

If $\rho$ is a density with $0 \leq \rho \leq 1$, mass 1 and compact support, then $E(\rho-\mu) \leq 0$ [25, Lemma I.1.8], i.e. $E(\rho) \leq 2\int U^\mu\,d\rho - E(\mu)$, so

$$E(\rho) + \int v_z\,d\rho \leq -E(\mu) + \lambda + \int (F_\mu - \lambda)\,d\rho. \tag{5.2}$$

The supremum of the left side over such $\rho$ supported in $[0,\infty)$ is the constrained equilibrium constant $C_z$; by (5.2), $C_z \leq \Phi_z(\mu,\lambda)$.

### 5.2. **The theorem.**

**Theorem 5.1** (Theorem B′)**.** *Let $0 < z \leq \sqrt{5}-2$ and $T = \chi_2(z)$. Let $\mu$ be admissible, $\lambda \in \mathbb{R}$ and $c_{\mathrm{cell}} \geq 0$, and suppose that*

(i) $F_\mu - \lambda \leq -1.13 - 0.94(x-4)$ *on* $[4,\infty)$, *and*
(ii) $F_\mu - \lambda \leq c_{\mathrm{cell}}$ *on* $[-1/100, 0]$.

*Then for every even $h \geq 100$,*

$$\log|\Delta_h(T)| \leq h^2\,\Phi_z(\mu,\lambda) + E_{\mathrm{an}}(h),$$

*where $c_1(h) := \log(2h) + 1 + \log(2+\log(2h+1))$ and*

$$E_{\mathrm{an}}(h) := 2h\log(4h) + h\log 4 + (2h+1)\log 3 + h\,c_1(h) + (c_{\mathrm{cell}} + 2.362)h - h\log h + 1.$$

*In particular* $\limsup_{h\to\infty} h^{-2}\log|\Delta_h(T)| \leq \Phi_z(\mu,\lambda)$.

For $m = 2$ the certified pair of Section 5.3 has $c_{\mathrm{cell}} = 5.38$, which gives the constant 7.742 of the earlier version. The proof follows the proof of Theorem B in [15] in the first two steps and then departs from it: the poles now lie on both sides of the atoms, which is handled by an exact Jensen-type bound (step 4), and the discretisation spreads the particles into cells (step 5) instead of using a lattice energy inequality. Steps 1–4 do not involve $z$; step 5 uses only (5.1), and step 6 only the hypotheses (i) and (ii).

*Proof.* Write $\phi := 1/D$.

*1. Node expansion.* For $a, b \in \mathbb{Q}[t]$, $(ab\phi)' = a'b\phi + ab'\phi + ab\phi'$, so $\Lambda(ab) = \sum_{t\ \mathrm{odd}\geq 1} (a,a')(t)\,M_t\,(b,b')(t)^{\mathsf{T}}$ with $M_t := z^t \left(\begin{smallmatrix} \phi' & \phi \\ \phi & 0 \end{smallmatrix}\right)(t)$. Let $H^{(t_{\max})}$ be the Hankel matrix of the sums truncated at $t \leq t_{\max}$. Each node $t$ carries a *value slot* (the row $(t^i)_{i<h}$) and a *derivative slot* (the row $(it^{i-1})_{i<h}$). By the Cauchy–Binet formula, $\det H^{(t_{\max})}$ is a sum over *patterns* $n = (n_t)$, $n_t \in \{0,1,2\}$, $\sum_t n_t = h$, and over pairs $(S,S')$ of slot sets with $n_t$ slots at node $t$, of $\pm \det E_S \det \mathbb{M}_{S,S'} \det E_{S'}$, where $E_S$ is the matrix of the rows in $S$ and $\mathbb{M}$ is block diagonal with blocks $M_t$. There are at most $4^h$ pairs $(S,S')$ per pattern.

*2. Slot weights.* $|\det \mathbb{M}_{S,S'}| \leq \prod_t (z^t\omega_t)^{n_t}$ with $\omega_t := \max(|\phi(t)|, |\phi'(t)|)$ (for $n_t = 2$ note $|\det M_t| = z^{2t}\phi(t)^2$). The $K$ numbers $t - x_k$ are distinct odd integers, so $|\phi'/\phi|(t) \leq \sum_k 1/|t-x_k| \leq 2\sum_{c=0}^{h} 1/(2c+1) \leq 2 + \log(2h+1)$, and $\omega_t \leq |\phi(t)|(2+\log(2h+1))$.

*3. Vandermonde determinants with derivative rows.* $|\det E_S| \le (4h)^h \prod_{t<t'} |t-t'|^{n_t n_{t'}}$. Indeed, replace the derivative-only slots $s_1, \dots, s_r$ of $S$ by value rows at free variables; the result is a confluent Vandermonde determinant $\pm \prod_{a<b}(a-b)^{n_a n_b}$, and $\det E_S$ is obtained by applying $\partial_{s_1} \cdots \partial_{s_r}$ and evaluating at the nodes. By induction, after $c$ derivatives the expression is a sum of at most $(h-1)^c$ terms, each a product of $c$ exponents (each $\le 2$) times $\prod (a-b)^{e'_{ab}}$ with $0 \le e'_{ab} \le n_a n_b$. At the nodes $|a-b| \ge 2 > 1$, so each term is at most $2^c \prod |a-b|^{n_a n_b}$, and $(2(h-1))^r \le (4h)^h$.

*4. The pole product.* For odd $t$ the numbers $t - x_k$ are the odd integers in $[t-h, t+h]$. For every odd $u$, $\log|u| \ge \frac{1}{2}\int_{u-1}^{u+1} \log|s|\, ds$, by concavity of log on $(0,\infty)$ and on $(-\infty, 0)$ (the singular endpoint is integrable). The intervals $[u-1, u+1]$ tile $[t-h-1, t+h+1]$, so with $x := t/h$ and $s = u/h$,

$$\begin{aligned}\log|D(t)| &\ge \tfrac{1}{2}\int_{t-h-1}^{t+h+1} \log|u|\, du \\ &= (h+1)\log h + h\Psi(x) + \tfrac{h}{2}\Big[\int_{x+1}^{x+1+1/h} + \int_{x-1-1/h}^{x-1}\Big] \log|s|\, ds.\end{aligned}$$

Each bracketed integral is over an interval of length $1/h$, hence at least $\int_{-1/(2h)}^{1/(2h)} \log|s|\, ds = -(\log(2h)+1)/h$. So $\log|D(t)| \ge (h+1)\log h + h\Psi(x) - \log(2h) - 1$ and, with step 2,

$$t\log z + \log\omega_t \le h\, v_z(t/h) - (h+1)\log h + c_1(h). \tag{5.3}$$

By steps 1–4, the pattern $n$ contributes at most $4^h (4h)^{2h} e^{D(n)}$ with $D(n) := \sum_{t\ne t'} n_t n_{t'} \log|t-t'| + \sum_t n_t\big(h\, v_z(t/h) - (h+1)\log h + c_1(h)\big)$.

*5. Discrete to continuum.* Spread the $n_t$ slots of node $t$ uniformly over $I_t := [(t-2)/h, t/h]$: the density $\rho_n := \sum_t \frac{n_t}{2}\mathbf{1}_{I_t}$ satisfies $0 \le \rho_n \le 1$, has mass 1, and is supported in $[-1/h, \infty)$. By (5.1), $v_z(t/h) \le v_z$ on $I_t$, so $\sum_t n_t h\, v_z(t/h) \le h^2 \int v_z\, d\rho_n$. For $t \ne t'$ with $|t-t'| = 2d$, writing $x = (t-2+2a)/h$, $a \in [0,1]$,

$$h^2 \iint_{I_t \times I_{t'}} \log|x-s|\, d\rho_n\, d\rho_n = n_t n_{t'}\big(\log|t-t'| - \log h - \mathrm{gap}(d)\big),$$

$$\mathrm{gap}(d) := \log d - \int_0^1\int_0^1 \log|d + a - a'|\, da\, da',$$

and $0 \le \mathrm{gap}(d) \le 0.1137/d^2$ (with equality at $d = 1$ up to rounding). A cell with itself gives $n_t^2(\log 2 - \log h - \frac{3}{2})$. Summing, and using $n_{t'} \le 2$ and $\sum_t n_t^2 \le 2h$,

$$\sum_{t\ne t'} n_t n_{t'} \log|t-t'| \le h^2 E(\rho_n) + h^2 \log h + 0.748h + 1.614h$$

($4\cdot 0.1137\,\zeta(2) \leq 0.748$ and $2(\frac{3}{2}-\log 2) \leq 1.614$). The terms in $\log h$ combine to $h^2\log h - h(h+1)\log h = -h\log h$, so

$$D(n) \leq h^2 J(\rho_n) - h\log h + h\,c_1(h) + 2.362h, \qquad J(\rho) := E(\rho) + \int v_z\,d\rho.$$

*6. Dual, the cell below* 0*, and far nodes.* By (5.2), $h^2 J(\rho_n) \leq h^2(-E(\mu)+\lambda) + h^2\int (F_\mu - \lambda)\,d\rho_n$. On $[0,4]$ the integral is at most $\int_0^4 (F_\mu-\lambda)_+ dx$, since $\rho_n \leq 1$. On $[-1/h, 0] \subset [-1/100, 0]$, which only the cell $I_1$ meets, (ii) gives $F_\mu - \lambda \leq c_{\mathrm{cell}}$, so the contribution to $h^2 J$ is at most $c_{\mathrm{cell}}h$. On $[4,\infty)$, by (i), a slot at a node $t \geq 4h+3$ contributes at most $-a_t$ with $a_t := h(1.13 + 0.94((t-2)/h - 4)) \geq 1.13h$; the node $4h+1$, whose cell straddles 4, has $F_\mu - \lambda < 0$ on its part beyond 4. So for every pattern

$$h^2 J(\rho_n) \leq h^2\Phi_z(\mu,\lambda) + c_{\mathrm{cell}}h - \sum_{t\geq 4h+3} n_t a_t.$$

The $2h+1$ odd nodes $t \leq 4h+1$ carry at most $3^{2h+1}$ patterns, and the nodes beyond contribute at most $\prod_{t\geq 4h+3}(1 + e^{-a_t} + e^{-2a_t}) \leq \exp\bigl(2e^{-1.13h}/(1 - e^{-1.88})\bigr) \leq e$. Collecting,

$$\begin{aligned}\log|\det H^{(t_{\max})}| \leq\ & h\log 4 + 2h\log(4h) + (2h+1)\log 3 \\ & + h^2\Phi_z(\mu,\lambda) + c_{\mathrm{cell}}h - h\log h + h\,c_1(h) + 2.362h + 1,\end{aligned}$$

uniformly in $t_{\max}$. As $t_{\max}\to\infty$ the truncated matrices converge entrywise to $(\Lambda(t^{i+j}))$, and the determinant is continuous. □

*Remark* 5.2. On $[-1/h, 0]$ we have $U^\mu(x) \leq \log(2+1/h)$ (as $\operatorname{supp}\mu \subset [0,2]$) and $v_z(x) \leq |\log z|/h - \Psi(0) = 1 + |\log z|/h$; so (ii) holds with $c_{\mathrm{cell}} = 2\log(2.01) + 1 + |\log z|/100 - \lambda$ whenever this is $\geq 0$. For any $h \geq 2$ one may instead use $U^\mu(x) \leq \int_1^2 \log(s+1)\,ds = 0.9095\ldots$ on $[-1/h, 0]$ (because $\mu \leq 1$ and $\operatorname{supp}\mu \subset [0,2]$); at $m=2$ this gives $F_\mu - \lambda \leq 7.23$ there and the bound of Theorem 5.1 for every even $h \geq 2$ with $c_{\mathrm{cell}} + 2.362$ replaced by 9.592. Only the $O(h)$ term changes.

### 5.3. The certificate at $\sqrt{5}-2$.

**Theorem 5.3.** *For $z = z_2 = \sqrt{5}-2$ there is an admissible pair $(\mu_U, \lambda_U)$ satisfying the hypotheses of Theorem 5.1 with $c_{\mathrm{cell}} = 5.38$ and*

$$\Phi_{z_2}(\mu_U, \lambda_U) \leq \Phi_U := -1.3592161834586374.$$

The proof is a finite computation in ball arithmetic (Arb [12], through python-flint [28]) at 128 bits. The density $\mu$ is constant on $N$ equal cells of $[0,2]$, with rational heights in $[0,1]$ and mass exactly 1 (checked in exact rational arithmetic). The energy $E(\mu)$ is computed from the exact double antiderivative $\frac{1}{2}u^2\log|u| - \frac{3}{4}u^2$ as a Toeplitz sum over cell offsets. The grid has step $d/r$ ($d = 2/N$ the cell width, $r$ even, so cell midpoints are grid points), and on each grid interval $I = [a,b]$:

- the three cells nearest to $I$ are treated exactly: each cell potential is unimodal with its minimum at the cell midpoint, so its maximum on $I$ is at an endpoint that can be chosen exactly;
- the far field $f = 2\sum_{\text{far}} \mu_j U_j$ (every far cell is at distance $\geq d$ from $I$) satisfies $f'' \geq -4/d$, because $f'' = -2\int_{\text{far}} \mu(s)(x-s)^{-2}ds$ and $\mu \leq 1$; and $v''(x) = 1/(x^2-1)$ is $\geq -1/(1-b^2)$ for $b < 1$ and $\geq 0$ for $a > 1$. A function $g$ with $g'' \geq -\kappa$ on $I$ satisfies $\sup_I g \leq \max(g(a), g(b)) + \kappa(b-a)^2/8$; this is applied to $f + v$ with $\kappa = 4/d + 1/(1-b^2)$ (if $b < 1$) or $\kappa = 4/d$ (if $a > 1$). On the two intervals touching $x = 1$, $v$ is bounded by $v(a)$;
- $\int (F_\mu - \lambda)_+ \leq \sum_I |I| \max(S_I - \lambda, 0)$, with $S_I$ the resulting upper bound on $I$; on $[4, \infty)$, $F_\mu(x) \leq 2\log x + x \log z - \log(x-1)$, which is $< \lambda$ and decreasing there.

A first version with per-cell maxima everywhere loses the cancellation between neighbouring cells and gives only $-1.313$ at $N = 100$. Table 2 lists the results; the last row proves Theorem 5.3.

| $N$ | $r$ | $-E(\mu)$ | $\lambda$ | $\int (F_\mu - \lambda)_+ \leq$ | certified $\Phi$ |
|---|---|---|---|---|---|
| 200 | 16 | 1.21902586 | $-2.96219282$ | 0.38673 | $-1.35643705$ |
| 400 | 32 | 1.21902867 | $-2.96297388$ | 0.38528 | $-1.35866119$ |
| 800 | 32 | 1.21902407 | $-2.96310589$ | 0.38505 | $-1.35903096$ |
| 1600 | 32 | 1.21902464 | $-2.96316943$ | 0.384929 | $-1.35921618345864$ |

TABLE 2. Dual certificates for Theorem 5.1 at $z = \sqrt{5} - 2$ (decimal values rounded; $\lambda$ is a rational number). For $N = 1600$ the certified Arb ball is $[-1.35921618345864 \pm 2.6 \cdot 10^{-15}]$, with upper endpoint $-1.3592161834586374$.

The side conditions of Theorem 5.1 hold for the certified pair: at $x = 4$, $2\log x + x \log z - \log(x-1) - \lambda = -1.137\ldots$, and the slope of this bound on $[4, \infty)$ is at most $2/x + \log z \leq -0.94$; on $[-1/h, 0]$, $h \geq 100$, we have $U^\mu \leq \log(2 + 1/h)$ and $v_z \leq 1 + 1.4437/h$, hence $F_\mu \leq 2.4108$ and $F_\mu - \lambda \leq 2.4108 + 2.96317 \leq 5.374 < 5.38$.

*Remark* 5.4 (the equilibrium at $m = 2$). An active-set quadratic program (and two independent codes) give the primal value $C_{z_2} \approx -1.3594016$, with maximiser saturated ($\rho = 1$) on $[0, 0.5475]$, a free band up to 1.9525, and $\lambda^* \approx -2.96317$; so the certificate is within about $1.9 \cdot 10^{-4}$ of the optimum. The finite-$h$ values $h^{-2}\log|\Delta_h(T)|$, computed in ball arithmetic, are $-1.4379$, $-1.4195$, $-1.4093$, $-1.4015$, $-1.3930$, $-1.3876$, $-1.3815$, $-1.3781$ at $h = 16$, 32, 48, 64, 96, 128, 192, 256. They increase towards the constant from below; fits of the form $a + b\log h/h\,(+c/h)$ give limits between $-1.373$ and $-1.364$, below $C_{z_2}$, so these data do not indicate that Theorem 5.1 is sharp. This costs nothing, since only the upper bound is used.

5.4. **The shift lemma.** A smaller $z$ only helps, and by a computable amount.

**Lemma 5.5** (shift lemma)**.** *Let $0 < z' \leq z \leq \sqrt{5}-2$ and $L := \log(z/z') \geq 0$. For every admissible $\mu$ and every $\lambda \in \mathbb{R}$,*

$$\Phi_{z'}(\mu, \lambda - L) \leq \Phi_z(\mu, \lambda) - \tfrac{L}{2}.$$

*If $(\mu, \lambda)$ satisfies (i) of Theorem 5.1 at $z$, then $(\mu, \lambda - L)$ satisfies (i) at $z'$; if $(\mu, \lambda)$ satisfies (ii) at $z$ with $c_{\mathrm{cell}}$, then $(\mu, \lambda - L)$ satisfies (ii) at $z'$ with $c_{\mathrm{cell}} + 1.01L$.*

*Proof.* Write $F_\mu$ and $F'_\mu$ for the functions at $z$ and $z'$. Since $v_{z'}(x) = v_z(x) - Lx$, we have $F'_\mu = F_\mu - Lx$ and

$$F'_\mu - (\lambda - L) = (F_\mu - \lambda) - L(x-1).$$

With $a := F_\mu(x) - \lambda$: on $[0,1]$, $(a + L(1-x))_+ \leq a_+ + L(1-x)$; on $[1,\infty)$, $(a - L(x-1))_+ \leq a_+$. Integrating, $\int_0^\infty (F'_\mu - \lambda + L)_+ \, dx \leq \int_0^\infty (F_\mu - \lambda)_+ \, dx + \frac{L}{2}$, so $\Phi_{z'}(\mu, \lambda - L) \leq -E(\mu) + \lambda - L + \int (F_\mu - \lambda)_+ + \frac{L}{2} = \Phi_z(\mu, \lambda) - \frac{L}{2}$. On $[4, \infty)$ the shift $-L(x-1)$ only lowers $F - \lambda$; on $[-1/100, 0]$ it raises it by at most $L(1 + 1/100)$. □

**Corollary 5.6.** *For every $m \geq 2$, the pair $(\mu_U, \lambda_U - L(m))$ satisfies the hypotheses of Theorem 5.1 at $z = z_m$ with $c_{\mathrm{cell}} = 5.38 + 1.01L(m)$, and*

$$\Phi_{z_m}\big(\mu_U, \lambda_U - L(m)\big) \leq \Phi_U - \tfrac{1}{2}L(m), \qquad L(m) := \log \frac{m + \sqrt{m^2+1}}{2 + \sqrt{5}}.$$

*In particular $\limsup_h h^{-2} \log |\Delta_h(\chi_2(z_m))| \leq \Phi_U - \frac{1}{2}L(m)$.*

*Proof.* $1/z_m = m + \sqrt{m^2+1}$, so $L(m) = \log(z_2/z_m) \geq 0$. Apply Lemma 5.5 to Theorem 5.3. □

For $m = 4$, $L(4) = 0.651077\ldots$ and the bound is $\Phi_{z_4} \leq -1.684755$. The proof of Theorem 1.1 uses only Corollary 5.6, i.e. Theorem 5.3 and the shift lemma.

5.5. **Direct certificates at $m = 4$ and $m = 8$.** The program of Section 5.3, with only $z$ changed, gives direct certificates at $z_4 = \sqrt{17}-4$ and $z_8 = \sqrt{65}-8$ (Table 3). They improve the analytic bound at these points but are not needed for Theorem 1.1, because the margins from Corollary 5.6 are already negative there (Table 4).

Their side conditions hold: at $x = 4$, $F_\mu - \lambda \leq -2.987$ ($m = 4$) and $-4.990$ ($m = 8$), with slopes at most $-1.595$ and $-2.276$; on the cell, $F_\mu - \lambda \leq 6.14$ ($m = 4$, $h \geq 100$). An independent projected-gradient computation of the primal problem gives $C_{z_4} \geq -1.7181612$ and $C_{z_8} \geq -2.0725380$, so the certificates are within about $2.1 \cdot 10^{-4}$ and $2.3 \cdot 10^{-4}$ of the optimum.

| $m$ | $N$ | $-E(\mu)$ | $\lambda$ | $\int(F_\mu-\lambda)_+\le$ | certified $\Phi$ |
|---|---|---|---|---|---|
| 4 | 200 | 1.36321476 | −3.71770185 | 0.63797 | −1.71651998814 |
| 4 | 800 | 1.36321826 | −3.71810316 | 0.63714 | −1.71774955485 |
| 4 | 1600 | 1.36321769 | −3.71817249 | 0.636999 | −1.7179552750082241 |
| 8 | 1600 | 1.43562390 | −4.44201260 | 0.934075 | −2.0723141156759906 |

TABLE 3. Direct certificates at $m=4$ and $m=8$ (upper endpoints of Arb balls). Not used in the proof of Theorem 1.1.

## 6. Nonvanishing: Lemma P′

6.1. **Setting.** Let $\ell\ge 7$ be a prime with $\ell\nmid m$, $h:=3\ell-1$ (even) and $K=3\ell$. The classes modulo $\ell$ are $C_\gamma=\{\gamma,\gamma+\ell,\gamma+2\ell\}$, $\gamma<\ell$, all of size 3. Since $|k-i|<3\ell<\ell^2$ there is a single level, and $N(i)=v_\ell(D'(x_i))=2$ for every $i$. Let $\mathfrak{l}$ be a prime of $\mathcal{O}_{\mathbb{K}}$ above $\ell$, $v:=v_{\mathfrak{l}}$, and $\zeta:=z^\ell \bmod \mathfrak{l}$. If $\ell$ splits or ramifies in $\mathbb{K}$, the residue field is $\mathbb{F}_\ell$ and $\zeta=z$; if $\ell$ is inert, $\zeta=\sigma(z)=-1/z$. Since $\ell\nmid 2m$, the elements $z$, $y$, $1-y=2mz$, $1-z^{-2}=-2m/z$ and $1-\zeta^{\pm 2}$ are $\mathfrak{l}$-units. (For ramified $\ell$, $\ell\mid d_m$, so $z\equiv -m$, $z^2\equiv -1$ and $1-z^{\pm 2}\equiv 2$.) For $m=2$: $\ell$ splits if $\ell\equiv\pm 1 \bmod 5$ and is inert otherwise.

When $d_m$ is not squarefree, primes dividing the index $[\mathcal{O}_{\mathbb{K}}:\mathbb{Z}[\sqrt{d_m}]]$ (for example $\ell=13$ at $m=70$, $d_{70}=4901=13^2\cdot 29$) need no special care: every step below works with the prime $\mathfrak{l}$ of $\mathcal{O}_{\mathbb{K}}$ and valuations in $\mathcal{O}_{\mathbb{K}}$, never with $\mathbb{Z}[\sqrt{d_m}]$.

*Positions.* $x_{\gamma+r\ell}=x_\gamma+2r\ell$ with $x_\gamma=2\gamma-3\ell+1\in[-3\ell+1,-\ell-1]$: in each class the pole with $r=0$ is large and negative ($|x|>\ell$), the pole with $r=1$ is small ($|x|<\ell$), and the pole with $r=2$ is large and positive. For odd $t$ we have $\ell\mid t-x_k$ exactly for $k$ in one class $C_{\gamma(t)}$, and the odd $t\ge 1$ with $\gamma(t)=\gamma$ form the progression $t=x_\gamma+\ell Y$, $Y\in\{3,5,7,\dots\}$. Put $t_\gamma:=x_\gamma+3\ell$.

**Lemma 6.1** (Lemma P′)**.** *Let $\ell\ge 7$ be prime with $\ell\nmid m$, $h=3\ell-1$, and $c_j:=[X^j]\Delta_h$. Then $v_\ell(c_0)=-2K$ and $v_\ell(c_j)\ge -2K+2j$ for all $j$. Hence $\Delta_h(a/b)\ne 0$ for every $a/b\in\mathbb{Q}$ with $v_\ell(b)\le 1$.*

*Proof. 1. Lemma B.* $\ell^{2K}\Delta_h(X)=(-1)^{K(K-1)/2}u_\ell\,\mathbf{1}^{\mathsf{T}}\operatorname{adj}(G^f(X))\mathbf{1}$ with $u_\ell:=\ell^{2K}/\prod_i D'(x_i)$ an $\ell$-unit.

*2. The $X$-part.* $B(f_if_j)=0$ for $i\ne j$, because $f_if_j/D$ is a polynomial, and $B(f_i^2)=-D'(x_i)z^{x_i}$, of valuation exactly 2. So $G^f(X)=G^f(0)+X\operatorname{diag}(d_i)$ with $v(d_i)=2$.

*3. Integrality and block diagonality.* By the row claim (3.2) with $N(i)-2\operatorname{lev}^*(\ell)=0$ (valid since $\ell\nmid m$), $G^f(0)$ is $\mathfrak{l}$-integral. On each progression, $\mathcal{M}$ restricted to $\gamma$ is $z^{t_\gamma}\sum_c z^{2\ell c}F'(t_\gamma+2\ell c)=z^{t_\gamma}\sum_b\Delta^bG_\gamma(0)z^{2\ell b}/(1-z^{2\ell})^{b+1}$, $G_\gamma(c):=F'(t_\gamma+2\ell c)$, with $1-z^{2\ell}\equiv 1-\zeta^2$ a unit; so (3.1) holds progression by progression. Off the diagonal, $F=\prod_{k\ne i,j}(t-x_k)$ has on the progression $\gamma$ exactly $3-[\,\gamma=c(i)\,]-[\,\gamma=c(j)\,]$ factors divisible by $\ell$ ($c(i)$ is the class of $i$),

and $F'$ keeps all but one; so a unit is possible only if $c(i) = c(j) = \gamma$. On the diagonal, the polynomial part on a progression $\gamma \neq c(i)$ has valuation at least $3-1=2$: the class $\gamma$ has three members, all after the initial segment of the telescoping order, and the derivative loses at most one. Hence $G^f(0) \bmod \mathfrak{l}$ is block diagonal over the classes, each block receiving only its own progression and its own pole terms.

*4. The blocks.* Fix $\gamma$ with members $k_r = \gamma + r\ell$, put $W(t) := \prod_{k \notin C_\gamma}(t - x_k)$ and $U_\gamma := \prod_{k \notin C_\gamma}(x_\gamma - x_k)$, so $W(t) \equiv U_\gamma \bmod \ell$ on the progression. Off the diagonal ($r \neq s$, $q$ the third index) $F = W(t)(t - x_{k_q})$ and $F' \equiv W \equiv U_\gamma$. For the diagonal polynomial part, $Q = W \cdot (t + x_{k_r} - x_{k_a} - x_{k_b}) + p(x_{k_r})\, W[x_{k_r}, t]$ with $\{a, b\}$ the other indices, $p(t) = (t - x_{k_a})(t - x_{k_b})$ and $p(x_{k_r}) \equiv 0 \bmod \ell^2$; so $Q' \equiv U_\gamma$ too. In the progression sum, $\Delta^0 G_\gamma(0) \equiv U_\gamma$ and $\Delta^b G_\gamma(0) \equiv 0$ for $b \geq 1$, so it is $\equiv U_\gamma z^{t_\gamma}/(1 - z^{2\ell}) \equiv U_\gamma z^{x_\gamma}\zeta^3/(1 - \zeta^2)$. The pole terms at $X = 0$ are $D'(x) z^x P(x)$ with $\ell^2 P(x_{k_r}) \equiv \zeta, 0, -\zeta^{-1}$ for $r = 0, 1, 2$ (the only odd multiples of $\ell$ below $|x| \leq 3\ell - 1$ in absolute value are $\pm\ell$), $D'(x_{k_r})/\ell^2 \equiv U_\gamma \cdot (8, -4, 8)_r$, and $z^{x_{k_r}} \equiv z^{x_\gamma}\zeta^{2r}$. Hence

$$G^f(0)|_{C_\gamma} \equiv e_\gamma M, \qquad e_\gamma := U_\gamma z^{x_\gamma},$$
$$M := c\,\mathbf{1}\mathbf{1}^{\mathsf{T}} + \operatorname{diag}(8\zeta, 0, -8\zeta^3), \qquad c := \frac{\zeta^3}{1 - \zeta^2}.$$

*5. The bordered determinant.* For $M = \operatorname{diag}(d_0, 0, d_2) + c\mathbf{1}\mathbf{1}^{\mathsf{T}}$ we have $\det M = c\, d_0 d_2 = -64c\zeta^4$ and $\mathbf{1}^{\mathsf{T}} \operatorname{adj}(M)\mathbf{1} = d_0 d_2 = -64\zeta^4$, both units. For the block-diagonal reduction,

$$\mathbf{1}^{\mathsf{T}} \operatorname{adj}\Big(\bigoplus_\gamma e_\gamma M\Big)\mathbf{1} = (\det M)^{\ell-1}\, \mathbf{1}^{\mathsf{T}} \operatorname{adj}(M)\mathbf{1} \Big(\prod_\gamma e_\gamma^3\Big) \sum_\gamma e_\gamma^{-1}.$$

Modulo $\ell$, $U_\gamma \equiv \prod_{k \not\equiv \gamma} 2(\gamma - k) \equiv 2^{3(\ell-1)}((\ell-1)!)^3 \equiv -1$ for every $\gamma$ (Fermat, Wilson). So $\sum_\gamma e_\gamma^{-1} \equiv -z^{3\ell-1} \sum_{\gamma<\ell} z^{-2\gamma} = -z^{3\ell-1}(1 - z^{-2\ell})/(1 - z^{-2})$, and $1 - z^{-2\ell} \equiv 1 - \zeta^{-2}$ and $1 - z^{-2} = -2m/z$ are units. Hence $\mathbf{1}^{\mathsf{T}} \operatorname{adj}(G^f(0))\mathbf{1}$ is an $\mathfrak{l}$-unit, and $v_\ell(c_0) = -2K$ by step 1.

*6. The gap.* $[X^j]\mathbf{1}^{\mathsf{T}} \operatorname{adj}(G^f(X))\mathbf{1}$ is a sum of products containing $j$ factors $d_i$, so it has valuation $\geq 2j$. If $v_\ell(b) \leq 1$, the $j$-th term of $\Delta_h(a/b)$ has valuation $\geq -2K + 2j - j > -2K$ for $j \geq 1$. □

The only place where $m$ enters is the list of units in the setting, and they are units exactly when $\ell \nmid 2m$. So every prime $\ell \geq 7$ with $\ell \nmid m$ works, whether it splits, is inert or ramifies. In [15] the partial sums reach past $3\ell$ and the block constants are fixed rationals that must avoid $\ell$, so finitely many primes are excluded; here only $n = \pm\ell$ enters. At $m = 2$, steps 4–5 were checked entrywise modulo $\ell$ for $\ell = 7, 11, \ldots, 41$ (split and inert), and the conclusion on the exact $\Delta_h$ for $\ell = 7, \ldots, 29$. For other $m$ the conclusion ($v_\ell(c_0) = -2K$ and the gap) was checked on the exact $\Delta_h$ at $m = 4$ ($\ell = 7, 11, 13, 17, 19$; $\ell = 17$ ramified), $m = 6$ ($\ell = 7, 11, 13$), and, by an independent implementation, at $m = 3, 7, 9, 11, 12, 13, 18, 21$ (17 cases,

among them three ramified $\ell$: $\ell = 17$ at $m = 13$, $\ell = 13$ at $m = 21$, and $\ell = 13$ at $m = 18$, where $d_{18} = 325 = 5^2 \cdot 13$). As a control, at $m = 7$, $\ell = 7$ (so $\ell \mid m$) the conclusion fails: $v_\ell(c_0) + 2K = -67$.

## 7. Proof of Theorem 1.1

7.1. **The proof.** Fix $m \geq 2$ with $m \notin \mathcal{E}$, let $\nu = 1 + v_2(m)$, and suppose that $T = \chi_2(z_m) = a/b$ with $a \in \mathbb{Z}$, $b \geq 1$. Let

$$\mathcal{H} := \{3\ell - 1 : \ \ell \geq 37 \text{ prime}, \ \ell \nmid m, \ \ell^2 \nmid b\},$$

an infinite set of even integers $h \geq 110$. For $h \in \mathcal{H}$, Lemma 6.1 gives $\Delta_h \not\equiv 0$ and $F_h(a/b) \neq 0$, where $F_h \in \mathbb{Z}[X]$ is the primitive polynomial of Section 2, of degree $\leq h$. We have $\log|F_h(T)| = \log|\Delta_h(T)| + \sum_p n_p \log p$, and along $\mathcal{H}$:

- odd primes $p \nmid m$: $\limsup h^{-2} \sum n_p \log p \leq \frac{4}{3}$ (Theorem 3.7);
- odd primes $p \mid m$, $e = v_p(m)$: $\limsup h^{-2} n_p \log p \leq c_p(e)$ (Lemma 3.8);
- $p = 2$: $\limsup h^{-2} n_2 \log 2 \leq c_2(\nu)$ (Proposition 4.2);
- size: $\limsup h^{-2} \log|\Delta_h(T)| \leq \Phi_U - \frac{1}{2} L(m)$ (Corollary 5.6).

So $\limsup_{h \in \mathcal{H}} h^{-2} \log|F_h(T)| \leq M(m)$, with $M(m)$ as in (1.2), and $M(m) < 0$ by Proposition 7.1 below. This contradicts Proposition 2.4. □

*Proof of Corollary 1.2.* At $m = 2$, $\nu = 2$, there are no odd primes dividing $m$, $L(2) = 0$, and $n_2 = O(h \log h)$ by Lemma 4.3. So $M(2) = \Phi_U + \frac{4}{3} = -0.0258828\ldots < 0$, and $T = \chi_2(\sqrt{5} - 2)$ is irrational; so is $24T = \pi^2 - 18\log^2\varphi$ by (1.1). □

7.2. **The margins.** Table 4 lists $M(m)$ for some $m$. Every entry is an upper bound computed in Arb.

**Proposition 7.1.** *$M(m) < 0$ for every integer $m \geq 2$ with $m \notin \mathcal{E}$, and $M(m) > 0$ for $m \in \mathcal{E} \cup \{1\}$.*

*Proof.* Write $m = 2^a q$ with $q$ odd. Since $m + \sqrt{m^2+1} > 2m$, $L(m) > \log(2m) - \log(2 + \sqrt{5})$. For odd $p$, $1/(4(p-1)) \leq 1/8$, so $\sum_{p^e \| q} c_p(e) \leq \frac{1}{4}\log q + \sum_{p \mid q} \frac{\log p}{4(p-1)} \leq \frac{3}{8}\log q$.

If $a \geq 1$, then $\nu = a + 1 \geq 2$ and $c_2(\nu) = (a-1)\frac{\log 2}{4}$, so

$$\begin{aligned} M(m) &< \left[\Phi_U + \tfrac{4}{3} + \tfrac{1}{2}\log(2+\sqrt{5}) - \tfrac{3}{4}\log 2\right] - \tfrac{a}{4}\log 2 - \tfrac{1}{8}\log q \\ &= 0.176075\ldots - 0.173287\ldots\, a - \tfrac{1}{8}\log q. \end{aligned}$$

This is negative unless $a = 1$ and $q = 1$, i.e. $m = 2$, where $M(2) = -0.025883$ exactly as computed above.

| $m$ | $\nu$ | $c_2(\nu)$ | odd $p \mid m$: $\sum c_p(e)$ | arithmetic | $\Phi_U - \frac{1}{2}L(m)$ | $M(m)$ |
|---|---|---|---|---|---|---|
| 2 | 2 | 0 | — | 1.333333 | −1.359216 | −0.025883 |
| 4 | 3 | 0.173287 | — | 1.506620 | −1.684755 | −0.178135 |
| 6 | 2 | 0 | 3: 0.411980 | 1.745313 | −1.883288 | −0.137975 |
| 8 | 4 | 0.346574 | — | 1.679907 | −2.025634 | −0.345727 |
| 10 | 2 | 0 | 5: 0.502949 | 1.836283 | −2.136509 | −0.300227 |
| 11 | 1 | 0.173287 | 11: 0.659421 | 2.166041 | −2.183949 | −0.017908 |
| 12 | 3 | 0.173287 | 3: 0.411980 | 1.918600 | −2.227291 | −0.308691 |
| 13 | 1 | 0.173287 | 13: 0.694674 | 2.201294 | −2.267184 | −0.065890 |
| 16 | 5 | 0.519860 | — | 1.853194 | −2.370753 | −0.517560 |
| 21 | 1 | 0.173287 | 3, 7: 0.979537 | 2.486157 | −2.506516 | −0.020359 |
| 3 | 1 | 0.173287 | 3: 0.411980 | | | +0.3720 |
| 5 | 1 | 0.173287 | 5: 0.502949 | | | +0.2160 |
| 7 | 1 | 0.173287 | 7: 0.567557 | | | +0.1147 |
| 9 | 1 | 0.173287 | 3 ($e = 2$): 0.686633 | | | +0.1091 |
| 15 | 1 | 0.173287 | 3, 5: 0.914929 | | | +0.0830 |

TABLE 4. Limit margins $M(m)$ per $h^2$ (upper bounds; the analytic column uses only Corollary 5.6). The rows below the line are the exceptional set $\mathcal{E}$, where the bound is positive. The direct certificates of Table 3 improve the rows $m = 4$ and $m = 8$ to $M(4) \leq -0.211335$ and $M(8) \leq -0.392407$.

If $a = 0$, then $\nu = 1$ and $c_2(1) = \frac{\log 2}{4}$, so

$$M(m) < \Big[\Phi_U + \tfrac{4}{3} + \tfrac{1}{2}\log(2+\sqrt{5}) - \tfrac{1}{4}\log 2\Big] - \tfrac{1}{4}\log q + \sum_{p \mid q} \frac{\log p}{4(p-1)}$$
$$\leq 0.522648\ldots - \tfrac{1}{8}\log q,$$

which is negative for $q > e^{8 \cdot 0.522648\ldots} = 65.443\ldots$. For the odd $m \leq 65$ we evaluate $M(m)$ in Arb, with the exact $L(m)$; for $3 \leq m \leq 65$ it is negative except for $m \in \mathcal{E}$, where the enclosures are positive, between +0.083 ($m = 15$) and +0.372 ($m = 3$) (Table 4). For $m = 1$ the independent interval computation mentioned below gives $M(1) \approx +0.4285$. □

So every $m > 65$ is covered by the closed form, the odd $m \leq 65$ by the table, and the set of $m \geq 2$ for which the method as written concludes is exactly $\{m \geq 2\} \setminus \mathcal{E}$. An independent computation in mpmath interval arithmetic (shift lemma only, no shared code) found $M(m) \geq 0$ exactly for $m \in \{1, 3, 5, 7, 9, 15\}$ among $1 \leq m \leq 3000$, and agreed with the table to all printed digits for $m \leq 60$. The tightest margins, $m = 11$ and $m = 21$, depend on the digits of $\Phi_U$ only at the level $10^{-2}$.

*The excluded values.* For $m \in \mathcal{E}$ the method does not conclude, although the exact finite-$h$ scores there are negative at $h \leq 48$ (Table 10). They rise with $h$ (for $m = 3$: $-0.1326$, $-0.0734$, $-0.0340$ at $h = 32, 48, 64$) and are not evidence for these $m$. For $m = 1$ the margin is positive and so are the

finite-$h$ scores: $+0.0595$, $+0.2144$, $+0.2524$, $+0.2876$ at $h = 16, 32, 48, 64$. So $\chi_2(\sqrt{2}-1) = \pi^2/16 - \frac{1}{4}\log^2(1+\sqrt{2})$ is out of reach of this method.

*Remark* 7.2 (towards $\mathcal{E}$). The following is not proved. The data after Lemma 3.8 suggest that the term $\frac{1}{4(p-1)}\log p$ in $c_p(e)$ can be removed by a plateau bound of the quality of Lemma 3.5 at special primes. That would bring $m = 9$ and $m = 15$ into Theorem 1.1. For $\nu = 1$ the exact 2-adic shares (for example 0.112 at $h = 64$, against the limit bound 0.173) suggest that $c_2(1) < \frac{1}{4}\log 2$; with direct certificates this might reach $m = 7$. The values $m = 3$ and $m = 5$ look out of reach of this machine.

### 7.3. **Effectivity.**

*Remark* 7.3 (effectivity at $m = 2$). Every bound above is explicit at finite $h$ except the prime number theorem in Theorem 3.7. At a single $h$ one can instead evaluate the proved finite-$h$ bound

$$\log|F_h(T)| \le h^2\Phi + E_{\mathrm{an}}(h) + B_{\mathrm{odd}}(h) + h(\lfloor \log_2 h \rfloor + 3)\log 2,$$

where $B_{\mathrm{odd}}(h) = \sum_{3\le p<K/3}(2h\,\mathrm{lev}^*(p) + v_p(h!))\log p + \sum_{K/3\le p<K}\min(2h + v_p(h!), 3(K-p))\log p$ is an exact finite sum. At $m = 2$, in exact arithmetic, this bound is negative at every even $h$ with $2432 \le h \le 40000$, and at every sampled $h$ up to $8.75\cdot 10^5$, with $|\mathrm{bound}|/h^2$ increasing towards 0.0259. A proof that it stays negative for all larger $h$ would need explicit error terms in the prime number theorem, which we have not carried out. As an effective consequence, if $\chi_2(\sqrt{5}-2) = a/b$ with $\log b \le 800$, a single $h = 3\ell - 1 \le 36000$ with $\ell \nmid b$ gives a contradiction. The exact scores $h^{-2}\log|F_h(T)|$ are much better than this bound, for example $-0.066$ at $h = 192$ (Table 9).

*Remark* 7.4 (effectivity at $m = 4$). The same finite-$h$ bound at $m = 4$, with the direct certificate of Table 3, Proposition 4.2 at $\nu = 3$ and Lemmas 3.2, 3.3, 3.5, is negative at every even $h$ with $178 \le h \le 40000$ and at every sampled $h$ up to $6\cdot 10^5$; the last nonnegative value is at $h = 176$. For example it is $-0.0749h^2$ at $h = 242$ and $-0.1648h^2$ at $h = 1004$. So if $\chi_2(\sqrt{17}-4) = a/b$, a single $h = 3\ell - 1$ with $\ell \nmid b$ and $h \ge 242$, of size about $5\log b + 300$, gives a contradiction. This uses the direct certificate, which the proof of Theorem 1.1 does not need.

## 8. Closed forms and known results

### 8.1. **Closed forms.**

Put $w_m := (1-z_m)/(1+z_m)$, so that Landen's identity reads

$$\chi_2(z_m) + \chi_2(w_m) = \frac{\pi^2}{8} - \frac{1}{2}\log z_m \,\log w_m. \tag{8.1}$$

For $m = 1$, $z_1 = w_1 = \sqrt{2}-1$, and (8.1) gives $\chi_2(\sqrt{2}-1) = \pi^2/16 - \frac{1}{4}\log^2(1+\sqrt{2})$. For $m = 2$, $w_2 = \varphi^{-1}$, and (8.1) with the value of $\chi_2(\varphi^{-1})$ gives (1.1).

**Proposition 8.1.** *Let $m \geq 1$. Then $w_m = (\sqrt{m^2+1}-1)/m$, $\sigma(w_m) = -1/w_m$ and $mw_m^2 + 2w_m - m = 0$; so $w_m$ is an algebraic integer if and only if $m \mid 2$. For $m \geq 3$, $z_m$ and $w_m$ are multiplicatively independent, and $z_m \wedge w_m \neq 0$ in $\Lambda^2(\mathbb{K}^\times) \otimes \mathbb{Q}$.*

*Proof.* The formula for $w_m$ and its quadratic equation are direct computations; the product of the roots is $-1$, so $\sigma(w_m) = -1/w_m$, and the monic equation $X^2 + (2/m)X - 1$ has integer coefficients iff $m \mid 2$. For $m \geq 3$, $w_m$ is not integral, so $v_\mathfrak{p}(w_m) < 0$ at some prime $\mathfrak{p}$, while $z_m$ is a unit. If $z_m^\alpha w_m^\beta = \pm 1$, then $\beta v_\mathfrak{p}(w_m) = 0$, so $\beta = 0$, and then $\alpha = 0$ since $0 < z_m < 1$. Hence $z_m$, $w_m$ are linearly independent in $\mathbb{K}^\times \otimes \mathbb{Q}$, and $z_m \wedge w_m \neq 0$. □

From $\delta \operatorname{Li}_2(x) = -(1-x) \wedge x$ one gets $\delta\chi_2(z) = \pm\frac{1}{2}\, w \wedge z$ modulo torsion, with $w = (1-z)/(1+z)$. The symbol of $\pi^2$ and of a product of two logarithms is 0 in $\Lambda^2$. So for $m \geq 3$ no dilogarithm identity that preserves the symbol expresses $\chi_2(z_m)$ through $\pi^2$ and products of logarithms; for $m = 1, 2$ the symbol vanishes, as Landen's evaluations require. This is a statement about symbols only: an unconditional statement that $\chi_2(z_m)$ is not of this form would need algebraic independence results that are not available.

By (8.1), Theorem 1.1 can equally be stated as $\chi_2(w_m) + \frac{1}{2}\log z_m \log w_m \notin \frac{\pi^2}{8} + \mathbb{Q}$. At $m = 4$, Hakimoglu-Brown [8, Cor. 14] (unrefereed) gives two identities for Rogers' dilogarithm $L$ over $\mathbb{Q}(\sqrt{17})$, in the bases $a = (\sqrt{17} - 3)/4$ and $b = \sqrt{17} - 4$. Their sum isolates $4\operatorname{Li}_2(b) - \operatorname{Li}_2(b^2) = 4\chi_2(b)$, and gives

$$12\chi_2(\sqrt{17}-4) = 18L(a) - 6L(a^2) + 18L(2a) - 3L(4a^2) - 2\pi^2 \\ - 6\log b \log(1-b) + 3\log b \log(1-b^2),$$

which we checked numerically to $10^{-120}$. It is a relation among non-elementary values ($a$ and $1-a$ are not units), not a closed form; Theorem 1.1 at $m = 4$ is equivalent to the irrationality of its right side.

8.2. **Known criteria.** Table 5 compares Theorem 1.1 with the general criteria of Section 1.1, evaluated at $z_m$ at the real place where $z_m$ is small. As there, $\chi_2(z) = z + \frac{z}{4}\Phi_2(\frac{1}{2}, z^2)$, and the height of $z_m^{-2}$ at that place is $\log(1/z_m)$. The evaluations are ours; that of [4, Thm. 1] through $\Phi_2(\frac{1}{2}, z^2)$ was re-derived from the published statement by the AI referee of Theorem 1.1. None of them has been checked by the authors of the criteria.

At $m = 4$ the first quantity is $-13.88$. The criterion of [4] gives the linear independence of $1$, $\Phi_1(\frac{1}{2}, z^2)$, $\Phi_2(\frac{1}{2}, z^2)$ over $\mathbb{K}$, hence $\chi_2(z_m) \notin \mathbb{K}$, a stronger conclusion than ours in its range. So the infinite tail of Theorem 1.1 is due to [4], and what Theorem 1.1 adds is the range $2 \leq m \leq 4\,346\,236$ with $m \notin \mathcal{E}$, by one argument for all $m$. As far as our searches show, no result in the literature covers any $m$ in this range; for $m \geq 3$ the searches were limited (Section 1.1), and a MathSciNet search and an expert query are still owed.

| criterion | key quantity at $z_m$ | covers |
|---|---|---|
| [4, Thm. 1], via $\Phi_2(\frac{1}{2}, z^2)$ | $\log(1/z_m) - 15.977968\ldots$ | $m \geq 4\,346\,237$, with $\chi_2(z_m) \notin \mathbb{K}$ |
| [4, Thm. 1], via $\mathrm{Li}_2(\pm z)$ | $\frac{1}{2}\log(1/z_m) - 15.78$ | $m \gtrsim e^{31.5}/2$ |
| [3] | weaker than [4] | nothing more |
| [13, Thm. 2.2] (unrefereed) | $\frac{1}{2}\log(1/z_m) - 26.84$ | $m \gtrsim e^{53.7}/2$ |
| Bombieri–André via [21] (unrefereed) | height $\frac{1}{2}\log(1/z_m)$ against about 1450 | $m \gtrsim e^{2900}$ |
| results at rational points | — | none |

TABLE 5. General criteria at $z_m$ (our evaluations).

## 9. Two points

9.1. **The statement.** Let $n \geq 6$ be even and $P_n(X) := X^4 + nX^3 + 2X^2 - nX + 1$. It is irreducible with four real roots: two in $(0,1)$, which we call $z > w$, and $-1/z$, $-1/w$; moreover $w = (1-z)/(1+z)$, and $z \approx 1 - 2/n$, $w \approx 1/n$. Let $\mathcal{K} := \mathbb{Q}(z)$, a totally real biquadratic field with $\mathrm{Gal}(\mathcal{K}/\mathbb{Q}) = \{1, s, l, sl\}$, $s(z) = -1/z$, $l(z) = w$. Put $u := z - 1/z$; then $u^2 + nu + 4 = 0$, and $\mathcal{F} := \mathcal{K}^s = \mathbb{Q}(u) = \mathbb{Q}(\sqrt{n^2-16})$, on which $l$ acts as the nontrivial automorphism, with $l(u) = 4/u$. With $\varepsilon_k := (k + \sqrt{k^2-4})/2$ we have $zw = \varepsilon_{n+2}^{-1}$ and $z/w = \varepsilon_{n-2}$. Put $T_z := \chi_2(z)$, $T_w := \chi_2(w)$; Landen's identity gives

$$T_n := T_z + T_w = \frac{\pi^2}{8} - \frac{1}{2}\log z \log w = \frac{\pi^2}{8} - \frac{1}{8}\log^2 \varepsilon_{n+2} + \frac{1}{8}\log^2 \varepsilon_{n-2}.$$

**Definition 9.1.** M1($n$) is the statement: $T_n \notin \mathbb{Q}$, or $T_z - T_w \notin \sqrt{n^2-16}\,\mathbb{Q}$.

Equivalently, there is no $\alpha \in \mathcal{F}$ with $T_z = \alpha$ and $T_w = l(\alpha)$: such an $\alpha$ would have $\alpha + l(\alpha) \in \mathbb{Q}$ and $\alpha - l(\alpha) \in \sqrt{n^2-16}\,\mathbb{Q}$, and conversely. (Exchanging $z$ and $w$ only changes the sign of $T_z - T_w$.)

**Theorem 9.2** (Theorem M1)**.** M1($n$) *holds for every* $n \in \mathcal{N} := \{730, 740, 750, 760, 800, 1000\}$.

The proof is computer-assisted: for each $n \in \mathcal{N}$ separately it uses two dual certificates checked in interval arithmetic. It is a disjunction at six specific values of $n$, not a statement about a range of $n$, and it is not formalised.

*The case* $n = 730$ (Theorem 1.3). Here $z = 0.99726400650629232574\ldots$, $w = 0.00136987072554999983 52\ldots$, $\varepsilon_{732} = 366 + \sqrt{133955} = 1/(zw)$, $\varepsilon_{728} = 364 + 11\sqrt{1095} = z/w$ and $\sqrt{730^2-16} = 22\sqrt{1101}$. Numerically $T_{730} = 1.2246689334990568135\ldots$, $T_z = 1.2232990624878816026\ldots$ and $T_w = 0.0013698710111752109\ldots$. The fields are $\mathcal{K} = \mathbb{Q}(\sqrt{133955}, \sqrt{1095})$ and $\mathcal{F} = \mathbb{Q}(\sqrt{1101})$, and the primes ramified in $\mathcal{K}$ are $2, 3, 5, 73, 367$. The other instances are in Table 6; in each case $\mathcal{K} = \mathbb{Q}(\sqrt{d_+}, \sqrt{d_-})$ with $d_\pm$ the squarefree parts of $(n \pm 2)^2 - 4$.

| $n$ | $\varepsilon_{n+2} = 1/(zw)$ | $\varepsilon_{n-2} = z/w$ | $\sqrt{n^2-16}$ | ramified in $\mathcal{K}$ |
|---|---|---|---|---|
| 730 | $366+\sqrt{133955}$ | $364+11\sqrt{1095}$ | $22\sqrt{1101}$ | $2, 3, 5, 73, 367$ |
| 740 | $371+2\sqrt{34410}$ | $369+4\sqrt{8510}$ | $16\sqrt{2139}$ | $2, 3, 5, 23, 31, 37$ |
| 750 | $376+5\sqrt{5655}$ | $374+5\sqrt{5595}$ | $2\sqrt{140621}$ | $2, 3, 5, 13, 29, 373$ |
| 760 | $381+2\sqrt{36290}$ | $379+6\sqrt{3990}$ | $12\sqrt{4011}$ | $2, 3, 5, 7, 19, 191$ |
| 800 | $401+20\sqrt{402}$ | $399+20\sqrt{398}$ | $4\sqrt{39999}$ | $2, 3, 67, 199$ |
| 1000 | $501+10\sqrt{2510}$ | $499+10\sqrt{2490}$ | $4\sqrt{62499}$ | $2, 3, 5, 83, 251$ |

TABLE 6. The six instances of Theorem 9.2.

9.2. **What changes at two points.** The machine is that of Section 2 with $z$ the root above: the same poles, atoms and functional, now with $A, B\colon \mathbb{Q}[t] \to \mathcal{K}$ and $\Delta_h \in \mathcal{K}[X]$. We write $y = z^2$; then $1 - y = -uz$. Since $u$ is an algebraic integer with $u\,l(u) = 4$, $u$ divides 4, and $z$, $w$ are units of $\mathcal{K}$ (roots of a monic polynomial with constant term 1).

*Reflection.* Lemma 2.2 holds with $\sigma = s$, the generator of $\mathrm{Gal}(\mathcal{K}/\mathcal{F})$: its proof used only $\sigma(z) = -1/z$ and that the coefficients are rational functions of $z$ with rational coefficients. So the odd-index entries lie in $e\,\mathcal{F}[X]$ with $e := z + 1/z$, $s(e) = -e$, $e^2 = u^2 + 4 = -nu \in \mathcal{F}$, and the rescaled matrix $E_{i+j}e^{[i \text{ odd}]-[j \text{ odd}]}$ shows $\Delta_h \in \mathcal{F}[X]$. The reflection removes the conjugates $-1/z$ and $-1/w$, where the series diverge.

*The second place.* The remaining conjugate, $l$, is not removed but convergent: the coefficients of $A$ and $B$ are rational functions of $z$, so $l(\Delta_h)(T_w) = \det(\Lambda_w(t^{i+j}))$, the Hankel determinant of the same machine at the weight $w$ (the series converge since $0 < w < 1$). The two embeddings of $\mathcal{F}$ thus see the $z$-machine and the $w$-machine, and the endgame uses the norm $\mathcal{F} \to \mathbb{Q}$.

*Odd primes.* At every prime of $\mathcal{K}$ above an odd $p$, $z$, $y$ and $1 - y = -uz$ are units. These are the only properties of $z$ that Lemmas 3.1–3.5 and Theorem 3.7 use, so they hold verbatim, with the valuation normalised by $v(p) = 1$; there are no special odd primes, including the ramified ones and those dividing $n^2 - 16$. The bound is the same at every prime $\mathfrak{p}$ of $\mathcal{F}$ above $p$.

*The prime* 2. Here Lemma 4.3 fails: at $n = 400$, $h = 48$, the exact 2-adic content is 388.5, above $h\lfloor \log_2 h \rfloor + 3h = 384$. It is replaced by the following weaker bound, which is the analogue over $\mathcal{K}$ of the case $\nu = 1$ of Proposition 4.2.

**Lemma 9.3** (Lemma 10″)**.** *Let $n$ be even and $\mathfrak{P}$ a prime of $\mathcal{K}$ above 2, with $v = v_{\mathfrak{P}}$ normalised by $v(2) = 1$. Then $-v(\Delta_h) \leq (h-2)^2/4 + 3h$.*

*Proof.* Since $n$ is even, $u/2$ is a root of $X^2 + (n/2)X + 1$, hence a unit; so $\nu := v(1 - y) = v(u) = 1$, instead of 2. Follow the proof of Lemma 4.1, which up to the valuations does not involve $z$. In (L), $y^b/(1-y)^{b+1}$ now has valuation $-(b+1)$, so $v(G_{ij}) \geq 1 - \pi + \min_{b<\pi}(v_2(b!) - 1) = -\pi$. In (U), with $v(\eta) = 1$, the first part has valuation $2 + s_2(N_0)$ and the second part valuation

$\geq 1 + s_2(N_0)$, so $v(G_{ij}) \geq 2$ for $N_0 \geq 0$; the bound no longer grows with $N_0$. Hence $v(G_{ij}) \geq -f(i+j)$ with $f(s) := (s-K)_+$, a convex function, and $v(\tilde{\Delta}_h) \geq -\max_\sigma \sum_i f(i+\sigma(i))$. If $i < i'$ and $\sigma(i) > \sigma(i')$, put $c = i + \sigma(i)$, $d = i' + \sigma(i')$, $a = i + \sigma(i')$, $b = i' + \sigma(i)$; then $a + b = c + d$ and $a < c, d < b$, so convexity gives $f(c) + f(d) \leq f(a) + f(b)$, and removing inversions does not decrease the sum. So the identity maximises, and $\sum_{i<h} f(2i) = 1 + 3 + \cdots + (h-3) = (h-2)^2/4$. Finally $-v(\Delta_h) = 3h - v(\tilde{\Delta}_h)$. □

So the prime 2 costs $\frac{1}{4}\log 2 = 0.1733\ldots$ per $h^2$ instead of 0. The mechanism that made it free in Lemma 4.3 is not available anywhere in the family:

**Proposition 9.4.** *For every $n$ there is a prime $\mathfrak{P}$ of $\mathcal{K}$ above 2 with $v_\mathfrak{P}(1-y) \leq 1$ (normalised by $v_\mathfrak{P}(2) = 1$). In particular $v_2(1-y) = 2$, which holds at $\sqrt{5} - 2$ and makes the prime 2 free there, cannot hold at every prime above 2.*

*Proof.* $z$ is a unit, so $v_\mathfrak{P}(1-y) = v_\mathfrak{P}(u)$. From $u\,l(u) = 4$ we get $v_\mathfrak{P}(u) + v_\mathfrak{P}(l(u)) = 2$, and $v_\mathfrak{P}(l(u)) = v_{l^{-1}(\mathfrak{P})}(u)$. If $v_\mathfrak{P}(u) \geq 2$ for every $\mathfrak{P} \mid 2$, then $v_\mathfrak{P}(l(u)) \leq 0$ for every $\mathfrak{P}$, so $v_{l^{-1}(\mathfrak{P})}(u) \leq 0$, a contradiction. □

For even $n$, $\nu = 1$ at every prime above 2; for odd $n$, $\nu \in \{0, 2\}$, and the prime with $\nu = 0$ is worse, so odd $n$ are not used. Proposition 9.4 does not show that the loss $\frac{1}{4}\log 2$ is necessary: exact profiles suggest that the true 2-adic cost is about $\frac{1}{8}\log 2$ per $h^2$, but we have not proved this.

*The analytic part.* Theorem 5.1 holds at a general weight $q \in \{z, w\}$, with $L := \log(1/q)$, the field $v_q(x) := -Lx - \Psi(x)$, and three changes. (a) $v_q$ is strictly decreasing on $(-\tanh L, \infty)$, which contains $[-1/h, \infty)$ if and only if $h > \coth L$; this is the range of $h$ ($\coth L_z = 364.9986\ldots$ at $n = 730$, and $\coth L_w < 1.0001$). (b) The dual density is only assumed $\geq 0$, of mass 1, with $\operatorname{supp}\mu \subset [0, X_e]$ for some $X_e \geq 4$ (at the $z$-place the support is long, $X_e \approx 1300$), and $\Phi_q(\mu, \lambda) := -E(\mu) + \lambda + \int_0^{X_e} (F_\mu - \lambda)_+\,dx$ with $F_\mu = 2U^\mu + v_q$. (c) The tail condition becomes: with $B(x) := 2\log x - \log(x-1) - Lx$, both $\beta := \lambda - B(X_e)$ and $\gamma := L - (X_e - 2)/(X_e(X_e-1))$ are positive. With $M_0 := 2\log(X_e+1) + 1 + L - \lambda$ and $\kappa := 2/(1 - e^{-2\gamma})$, the proof of Theorem 5.1 gives, for every even $h > \coth L$,

$$\log\bigl|\det(\Lambda_q(t^{i+j}))_{i,j<h}\bigr| \leq h^2\Phi_q(\mu,\lambda) + E_q(h),$$

$$E_q(h) := 2h\log(4h) + h\log 4 + (hX_e/2 + 2)\log 3 + h\,c_1(h) + 2.362h - h\log h + h\max(M_0, 0) + \kappa.$$

The pattern count below $X_e h$ replaces $(2h+1)\log 3$, the cell $[-1/h, 0]$ costs $h\max(M_0, 0)$ (here $U^\mu \leq \log(X_e+1)$ and $v_q \leq L+1$ there), and the far nodes cost $\kappa$. Since $E_q(h) = O(h\log h)$, the lim sup is again $\Phi_q$.

*Nonvanishing.* Lemma 6.1 holds over $\mathcal{F}$ for $h = 3\ell - 1$ with $\ell \geq 7$ prime and unramified in $\mathcal{K}$. Now $\zeta := z^\ell \bmod \mathfrak{l}$ is $\mathrm{Frob}_\mathfrak{l}(z)$, one of four values:

| $\mathrm{Frob}_{\mathfrak{l}}$ | 1 | $s$ | $l$ | $sl$ |
|---|---|---|---|---|
| $\zeta$ | $z$ | $-1/z$ | $w$ | $-1/w$ |
| $1-\zeta^2$ | $-uz$ | $u/z$ | $-(4/u)w$ | $(4/u)/w$ |
| $1-\zeta^{-2}$ | $u/z$ | $-uz$ | $(4/u)/w$ | $-(4/u)w$ |

Every entry is an $\mathfrak{l}$-unit because $u \mid 4$ and $\ell$ is odd, and steps 1–5 of Lemma 6.1 use only these units and $1-z^{-2} = u/z$. Since $\ell$ is unramified, $v_{\mathfrak{l}}$ restricts to $v_{\mathfrak{L}}$ on $\mathcal{F}$ for $\mathfrak{L} = \mathfrak{l} \cap \mathcal{F}$, and step 6 gives $\Delta_h(\alpha) \neq 0$ for every $\alpha \in \mathcal{F}$ with $v_{\mathfrak{L}}(\alpha) \geq -1$, whether $\ell$ is split or inert in $\mathcal{F}$.

**Lemma 9.5** (Lemma E′, the norm endgame)**.** *Let $n \geq 6$ be even, and suppose that for each $q \in \{z, w\}$ there is a pair $(\mu_q, \lambda_q)$ as in (b) above ($\mu_q \geq 0$, mass 1, $\operatorname{supp}\mu_q \subset [0, X_e]$ with $X_e \geq 4$; the cap $\mu_q \leq 1$ is not required) satisfying the tail condition (c), with*

$$M_1(n) := \tfrac{1}{2}\big(\Phi_z(\mu_z, \lambda_z) + \Phi_w(\mu_w, \lambda_w)\big) + \tfrac{4}{3} + \tfrac{1}{4}\log 2 < 0.$$

*Then* M1($n$) *holds.*

*Proof.* Suppose $\alpha \in \mathcal{F}$ has $T_z = \alpha$ and $T_w = l(\alpha)$, and fix $b \geq 1$ with $b\alpha \in \mathcal{O}_{\mathcal{F}}$. Let $\mathcal{H} := \{3\ell - 1 : \ell \geq 7 \text{ prime, unramified in } \mathcal{K},\ \ell \nmid b,\ 3\ell - 1 > \coth L_z\}$, an infinite set; note $\coth L_w < \coth L_z$. For $h \in \mathcal{H}$, Lemma P′ over $\mathcal{F}$ gives $\Delta_h(\alpha) \neq 0$. For a prime $\mathfrak{p}$ of $\mathcal{F}$ put $n_{\mathfrak{p}} := -v_{\mathfrak{p}}(\Delta_h)$, which may be a half-integer at ramified primes, and $n_p := \max(0, \max_{\mathfrak{p}\mid p}\lceil n_{\mathfrak{p}}\rceil)$, $c_h := \prod_p p^{n_p}$. Then $c_h\Delta_h \in \mathcal{O}_{\mathcal{F}}[X]$, and by the odd-prime lemmas and Lemma 9.3,

$$\log c_h \leq \sum_{3 \leq p < K} B_p \log p + \big((h-2)^2/4 + 3h\big)\log 2 + \theta(K) + \log 2,$$

where $B_p$ are the bounds used in Theorem 3.7; the term $\theta(K) = O(h)$ pays for all ceilings. So $\limsup_{h\in\mathcal{H}} h^{-2}\log c_h \leq \frac{4}{3} + \frac{1}{4}\log 2$. The algebraic integer $\beta := b^h c_h \Delta_h(\alpha)$ is nonzero. Its two real embeddings are $b^h c_h \Delta_h(T_z) = b^h c_h \det(\Lambda_z(t^{i+j}))$ and $b^h c_h\, l(\Delta_h)(T_w) = b^h c_h \det(\Lambda_w(t^{i+j}))$, so $N_{\mathcal{F}/\mathbb{Q}}(\beta) = b^{2h} c_h^2 \det(\Lambda_z(t^{i+j}))\det(\Lambda_w(t^{i+j}))$ is a nonzero integer. Taking logarithms, dividing by $2h^2$, and applying the analytic bound at both places, $0 \leq \frac{4}{3} + \frac{1}{4}\log 2 + \frac{1}{2}(\Phi_z + \Phi_w) = M_1(n) < 0$, a contradiction. □

9.3. **The certificates.** For each $n \in \mathcal{N}$ and $q \in \{z, w\}$ the certificate is a piecewise constant $\mu_q \geq 0$ on a grid of $[0, X_e]$ with mass exactly 1, a rational $\lambda_q$, a rigorous Arb upper bound for $\Phi_q(\mu_q, \lambda_q)$, and an Arb check of the tail condition with $X_e \geq 4$; the logarithm $L = \log(1/q)$ is an Arb ball computed from the quartic, so every bound encloses the value at the true $L$. Two independent evaluators were used, and where both certified the same $(n, q)$ the smaller bound is taken. Table 7 lists the results; $\frac{4}{3} + \frac{1}{4}\log 2 = 1.50662012847331966\ldots$, and every $M_1(n)$ is computed in Arb from the printed balls and lies entirely below 0.

| $n$ | $\Phi_z \le$ | $\Phi_w \le$ | $M_1(n) \le$ |
|---|---|---|---|
| 730 | 0.97551590832772 | −3.98960878950 | −0.000426312 |
| 740 | 0.98154355971108 | −3.99641267680 | −0.000814430 |
| 750 | 0.98568546476333 | −4.00312517276 | −0.002099725 |
| 760 | 0.98610919549783 | −4.00974867868 | −0.005199613 |
| 800 | 1.00091072617 | −4.03494034059 | −0.010394678 |
| 1000 | 1.05765433525 | −4.14651125316 | −0.037808330 |

TABLE 7. Certified bounds for Theorem 9.2 (upper endpoints of Arb balls, rounded up). At the $z$-place $X_e = 1300$ for $n \le 760$, and $X_e \approx 1202$ and 1503 for $n = 800$, 1000; at the $w$-place $X_e \approx 4$.

The constant $\Phi_z$ is positive because $z \approx 1-2/n$ gives the $z$-machine almost no decay; the $w$-place, with $w \approx 1/n$, pays for it. The side constants are comfortable: at $n = 730$, for example, $\beta = 0.5719$, $\gamma = 0.001971$, $M_0 \le 11.17$, $\kappa \le 508.4$ at the $z$-place, and $\beta = 16.41$, $\gamma = 6.426$ at the $w$-place. A third evaluator, written independently of both, confirms $M_1(n) < 0$ for all six $n$ and gives $M_1(730) \le -0.000452$. As a validation, the three codes applied at the weight $\sqrt{5} - 2$ of Corollary 1.2 give −1.35926599, −1.35939764 and −1.35939884, consistent with Theorem 5.3 and the primal value −1.3594016 of Remark 5.4.

9.4. **Scope.** We state the limits of Theorem 9.2 explicitly.

- It is a disjunction. We do not know which of the two statements holds, and neither is known by other means: $T_n \notin \mathbb{Q}$ is a statement of strong-four-exponentials type, and we know of no result on $T_z - T_w$.
- For very large $n$ a general criterion may apply: our evaluation of [4, Thm. 1] with the shift $\frac{1}{2}$ at $w^2$, at the real place where $w$ is small, suggests that its condition holds once $n \gtrsim 7.6 \cdot 10^{13}$, which would give $\chi_2(w) \notin \mathcal{K}$ and hence M1($n$) for those even $n$. This evaluation is a sketch that has not been checked in detail or by a referee. For $730 \le n \le 1000$ the same quantity lies between −12.7 and −12.5, so Theorem 9.2 is not implied.
- It is proved for the six listed values of $n$ only, each with its own certificates. Even $n$ between them are not covered, and odd $n$ are not covered at all: Lemma 9.3 needs $n$ even, and for odd $n$ one prime above 2 has $v(1 - y) = 0$.
- For all even $n \ge 730$ the statement is supported only numerically. A proof would need interval certificates in $n$ (the dual bound is monotone in $L$, but this needs to be written) and an argument as $n \to \infty$; neither exists.

- The method cannot reach $n = 720$: there the primal value of the equilibrium problem gives $M_1(720) \geq +0.00098$ (computed numerically, with a density admissible up to rounding of order $10^{-12}$), so no dual certificate can succeed. Numerically the threshold of the method lies in $(720, 730)$.
- The theorem is not formalised in Lean. Its proof was checked by two AI referees and has not been read by a human.

## 10. The formalisation

10.1. **What is formalised.** The Lean project [17] proves Theorem 1.1, Corollary 1.2 and the identity (1.1) in Lean 4 [6] (version v4.35.0) with Mathlib [29] (version v4.35.0 [30]). The machine is formalised once, with $m$ as a parameter. Theorem U, its instance $m = 2$, was proved first; Theorem V adds the ingredients that depend on $m$: Lemma 4.1, Lemma 3.8, the shift lemma, the margins, and Lemma 6.1 for every $m$. The project has 76 modules under `Chi2U/`, with about 20,100 lines, of which about 7,800 are ported verbatim from the formalisation of [15] [16]: the prime number theorem, the interval arithmetic, the generic analytic files, the criterion and the definition of $\chi_2$. All declarations are in the namespace `Chi2U`. The project is available at `https://github.com/telaviv/pi2-log2-golden-irrationality` [17], together with a copy of this paper and the scripts of Section 11. Theorem 9.2 is not formalised.

10.2. **The statement.** The definitions that the reader has to check are in `Defs.lean`, `Statement.lean`, `V/Defs.lean` and `Golden/Corollary.lean`:

```
noncomputable def chi2Term (z : ℝ) (n : ℕ) : ℝ :=
    z ^ (2 * n + 1) / (2 * (n : ℝ) + 1) ^ 2
noncomputable def chi2 (z : ℝ) : ℝ := ∑' n : ℕ, chi2Term
z n
def V.TheoremV : Prop := ∀ m : ℕ, 2 ≤ m → m ∉ ({3, 5, 7,
9, 15} : Finset ℕ) →
    Irrational (chi2 (Real.sqrt ((m : ℝ) ^ 2 + 1) - m))
theorem V.theoremV : V.TheoremV
def TheoremU : Prop := Irrational (chi2 (Real.sqrt 5 - 2))
theorem chi2_sqrt5_sub_two_irrational : TheoremU
theorem chi2_sqrt5_sub_two_eq :
    chi2 (Real.sqrt 5 - 2) = Real.pi ^ 2 / 24 - 3 / 4 *
Real.log goldenRatio ^ 2
theorem golden_irrational' :
    Irrational (Real.pi ^ 2 - 18 * Real.log goldenRatio ^ 2)
```

`V.TheoremV` and `V.theoremV` are in `V/Defs.lean` and `V/Interface.lean`. Of the other three theorems, the first is in `Main.lean`, the second in `Golden/ClosedForm.lean`, the third in `Golden/Corollary.lean`. The definition of $\chi_2$ is identical to that of [16], and `Defs.lean` proves summability for $|z| \leq 1$, so `chi2 (Real.sqrt 5 - 2)` and, since $0 < \sqrt{m^2+1} - m < 1$, `chi2 (Real.sqrt (m ^ 2 + 1) - m)` are the genuine sums of the series.

`goldenRatio` is Mathlib's $(1+\sqrt{5})/2$ and `Irrational` is Mathlib's predicate $x \notin \mathbb{Q}$. The identity (1.1) is proved from Landen's identity for $\chi_2$, Euler's reflection formula and Landen's identity for $\mathrm{Li}_2$ (`Golden/Landen.lean`, `Golden/Dilog.lean`), so the golden-ratio form is not a separate assumption.

The command `lake env lean Axioms.lean` prints

```
'Chi2U.V.theoremV' depends on axioms:
    [propext, Classical.choice, Quot.sound]
'Chi2U.chi2_sqrt5_sub_two_irrational' depends on axioms:
    [propext, Classical.choice, Quot.sound]
'Chi2U.golden_irrational'' depends on axioms:
    [propext, Classical.choice, Quot.sound]
```

and the same list for `chi2_sqrt5_sub_two_eq` and for each of the main ingredients listed in `Axioms.lean`. These are Lean's three standard axioms. The development contains no `sorry`, no added axiom, no `native_decide`, no `implemented_by` and no `@[extern]`, so no step relies on compiled code.

10.3. **Correspondence with the paper.** Table 8 lists the main formal counterparts. Paths are relative to `Chi2U/`. The machine files take the parameter $m$, called `mz` there (`Qd mz` = `QuadraticAlgebra` $\mathbb{Q}$ $(m^2+1)$ 0, `zK mz`, `Delta mz h`); the entries marked $m = 2$ are used only for Corollary 1.2.

10.4. **Differences from the written proof.** The formal proof establishes what the endgame needs, and in several places it takes a route that was shorter in Lean.

- *The machine.* $\Delta_h$ is *defined* as the determinant of the rational coordinates of the matrix $\tilde{E}$ of Corollary 2.3, so $\Delta_h \in \mathbb{Q}[X]$ holds by definition; Lemma 2.2 and Corollary 2.3 then identify it with the Hankel determinant over $\mathbb{K}$ and its value at $T$ with $\det(\Lambda(t^{i+j}))$.
- *Theorem B′.* The formal proof does not follow Section 5. It follows the lattice-energy route of the formal proof of Theorem B in [15]: the Andreief identity for the node expansion, the bound $h^h(1+2/h)^{h^2}$ for determinants with derivative rows via averages over roots of unity (instead of step 3), and a discrete Pólya-type lattice energy inequality comparing the patterns, which live on the odd nodes, with a reference sequence on all of $\mathbb{N}$ (instead of steps 5–6). The ingredient not present in [16] is the two-sided Jensen bound of step 4 (`log_abs_Dpoly_ge`), which gives the node weight (5.3). This route evaluates the dual at the nodes pointwise, so it never meets the cell $[-1/h, 0]$ and does not use the monotonicity (5.1) (which is nevertheless proved, as `vU_strictAntiOn`, on $[-\frac{1}{2}, \infty)$). It is stated asymptotically, like Theorem B of [15]: the explicit $E_{\mathrm{an}}(h)$ is not formalised.
- *The certificate.* The formal certificate (`DualCertU`) consists of a step density with 50 cells of $[0, 2]$ (49 nonzero), grid width $1/L$ with $L = 200$ on $[0, X]$ with $X = 4$, rational cell bounds $S_k$ for

| paper | Lean declaration (file) |
|---|---|
| $\mathbb{K}$, $z$, $\sigma$ | `Qd`, `zK`, `sig`, `emb` (`Field`); $m = 2$: `Q5 = Qd 2` |
| Lemma 2.1, (2.1) | `lemma1`, `aeval_chi2_Delta` (`PoleExpansion`); `Lmom_eq_mahler` (`Mahler`) |
| Lemma 2.2, Cor. 2.3 | `eval₂_Delta`, `aeval_Delta`, `Delta_map` (`LemmaR`) |
| Prop. 2.4 | `irrational_of_poly_approx` (`Criterion`); `MachineBoundsU`, `irrational_of_boundsU` (`MainCond`) |
| Lemma 3.3 | `AK_eq` (`Arith/ClassCount`) |
| Lemma 3.4 | `lemmaB` (`Arith/LemmaB`) |
| Lemmas 3.2, 3.5 | `lemma9` (`Arith/OddPrimes`, `Arith/Lemma9Levels`); `Arith/Lemma8One`; `padicNorm_Delta_odd` (`Arith/OddExp`) |
| Thm. 3.7 | `theoremA` (`Arith/PrimeSums`); `denAll_int`, `log_denAll_le` (`Arith/Assembly`) |
| Lemma 4.3 ($m = 2$) | `lemma10_L`, `lemma10_U`, `padicNorm_Delta_two` (`Arith/TwoAdic`) |
| Lemma 4.1 | `lemma10nu_proof` (`V/Lemma10nu`, `V/TwoAdicNu`); `lemma10nu_size` (`V/Defs`) |
| Lemma 3.8 | `lemmaS_proof` (`V/LemmaS`, `V/S*`); `AK_levels_size_proof` (`V/AKSize`) |
| Thm. 5.1 (other route) | `log_abs_Dpoly_ge`, `DualCertU`, `DualCertU.analytic` (`Analytic/TheoremBU`) |
| Thm. 5.3 (coarser) | `exists_certU`, `exists_certU'` (`Cert/CertU`); data `Cert/DataU` |
| Lemma 5.5 | `shift_lemma_proof` (`V/Shift`) |
| margins, Section 7.2 | `margin_V_proof` (`V/Margin`) |
| Lemma 6.1 | `lemmaP`, `nonvan_frequently` (`NonVan/LemmaP`); `lemmaP_m_proof` (`V/NonVan`) |
| Section 7 | `denV0_int`, `log_denV_le`, `exists_boundsV`, `theoremV` (`V/Interface`); $m = 2$: `boundsU`, `chi2_sqrt5_sub_two_irrational` (`Main`) |
| (1.1) | `chi2_landen`, `Li2_reflection`, `Li2_landen` (`Golden/Landen`); `chi2_sqrt5_sub_two_eq` (`Golden/ClosedForm`) |

TABLE 8. Formal counterparts of the main steps.

$2U^{\mu} + v$, and the tail conditions; its exponent is $-E(\mu) + \lambda + \frac{1}{L}\sum_k (S_k - \lambda)_+$. The cell bounds use chords and tangents of each term, as in [15], which keeps the cancellation between neighbouring cells without the near/far split of Section 5.3. The certified value is $\Phi_{\mathbb{Q}} = -54155169767979/40000000000000 \approx -1.3538792$, coarser than Theorem 5.3 but enough: the assembly (`boundsU`) uses $\beta = -27/20$, so the formal margin is $\frac{4}{3} - \frac{27}{20} = -\frac{1}{60}$.

- *Lemma 8′.* For the generic primes only the single-level range $K/3 \le p$ used by Theorem 3.7 is formalised, by a crude row bound on the Lagrange Gram matrix and Lemma 3.4 instead of the atom expansion. The multi-level statement needed by Lemma 3.8 is proved directly (`V/LemmaS` and its helper files) instead of through the expansion of Lemma 7′: a Schur-complement form of the determinant, a multilinear expansion of its bottom rows, and a Newton (divided-difference) factorisation that extracts the squared Vandermonde factor. The polynomial factor enters through integer row and column exponents $(i-J)_+$, $(j-J-1)_+$ with the same total $(J-1)^2$. This proof does not use $p \mid m$.
- *Lemma $10_\nu$.* The Karamata step is replaced by a weighted Leibniz bound: by concavity $\hat g(2i)+\hat g(2j) \le 2\hat g(i+j)$, so integer row weights $\lfloor \hat g(2i)/2 \rfloor$ and column weights $\lceil \hat g(2j)/2 \rceil$ give $v_2(\det G) \ge \sum_i \hat g(2i)$. Some entry bounds are coarser than in Lemma 4.1, where the slack of its final bound allows it. For odd $m$ no half-integral valuations occur: every bound is a rational power of 2 times an element of the order.
- *The shift lemma* is applied to the formal certificate of Theorem U, in its discrete format: each grid cell is split into $r$ subcells, and the left-endpoint sum loses $L(m)/(2rL)$, which is absorbed by taking $r$ large. So the formal analytic constant at $z_m$ is $\Phi_{\mathbb{Q}} - \frac{1}{2}L(m) + \delta$ for any $\delta > 0$, with $\Phi_{\mathbb{Q}} \approx -1.3538792$ in place of $\Phi(z_2)$ of Theorem 5.3.
- *The margins* are proved with this coarser constant, and the exception set is still exactly $\mathcal{E}$. Even $m \ge 4$ and odd $m \ge 69$ are handled by a closed form written with $\log m$ (the tightest case is $m = 69$, with slack about $10^{-3}$), $m = 2$ exactly, and the 28 odd $m \in [11, 67] \setminus \mathcal{E}$ one by one with rational bounds on the logarithms (for instance $\log(2+\sqrt{5}) \le \frac{25}{12}\log 2$).
- *Lemma P′.* Congruences are taken in a residue field of the order at a prime above $\ell$, with $\zeta := z^\ell$ itself; no identification of $\zeta$ by Frobenius is needed, only $1 - \bar z^{2\ell} = (2m\bar z)^\ell$. The formal statement needs only $\ell \nmid 2m$, for every $m$. The progression sums are handled by an exact shift identity instead of the Newton form.
- *The endgame* uses only even $h$, through $P_h := \Delta_{2\lfloor h/2 \rfloor}$ in the criterion of [16].

10.5. **How the certificate is checked.** The certificate data (`Cert/DataU.lean`) were produced by the script `scripts/gen_cert_u.py` (run as `python scripts/gen_cert_u.py 50 8 4 Chi2U/Cert/DataU.lean`). The script is not trusted: every check is a decidable statement about the rational data, proved by `decide +kernel` in `Cert/CertU.lean` in 16 chunks of grid cells, together with a table of 2004 logarithm enclosures, the energy bound and the tail. The dyadic enclosure of $\log 2$ used by the table is itself checked

by `decide +kernel`. A wrong value would make the build fail but could not make a false statement provable.

10.6. **Cost.** The kernel check of the certificate takes about 6.5 minutes. A build of a fresh clone of the repository, with Mathlib's prebuilt cache and at most four parallel Lean processes, took 9 minutes 35 seconds (9110 jobs, no errors) on a Windows 11 desktop with 32 logical processors and 31.7 GB of RAM, before the files for general $m$ were added; the complete project has 9124 jobs, and `Axioms.lean` reports only the three standard axioms.

10.7. **What has to be trusted.** Beyond the Lean kernel and the three axioms, the reader has to trust that the definitions displayed above express Theorem 1.1 and Corollary 1.2, and the Mathlib definitions they use (real numbers, `tsum`, `Real.sqrt`, `Real.log`, `Real.pi`, `goldenRatio`, `Irrational`). Nothing else, in particular no numerical data and no generating script, needs to be trusted. To avoid trusting the elaborator's handling of the environment, every declaration of the 76 project modules was replayed through the kernel with `leanchecker` (shipped with the Lean toolchain), each module against the compiled environment of its imports, on 8 November 2026; all modules passed, in about 52 minutes in total. Mathlib itself was not replayed from scratch (`--fresh`). Two AI referees, one for Corollary 1.2 and one for Theorem 1.1, checked the fidelity of the statements (the definitions above, traced to Mathlib, and the absence of hypotheses, custom axioms and compiler trust) and found them faithful; the second also confirmed the exception set $\mathcal{E}$ independently in interval arithmetic for $m \leq 3000$. Apart from the check by the Lean kernel, all verification of the formalisation so far was carried out by AI systems; no human expert has yet reviewed the Lean code.

## 11. Computations and reproducibility

11.1. **Software.** The computations use python-flint 0.9.0 [28]: exact arithmetic over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{d_m})$ for every statement about valuations, and Arb ball arithmetic [12] for every real number that feeds a claim. Hankel determinants are evaluated at $T$ in ball arithmetic with working precision about $3.6h^2 + 8000$ bits at $m = 2$, and at least $2 \cdot$ (bits of the coefficients) $+ 4000$ bits in the independent check for general $m$. The formalisation uses Lean v4.35.0 and Mathlib v4.35.0.

11.2. **What each computation is used for.**

- *Proof inputs.* The certificate of Theorem 5.3 (`cert_U.py`, run with arguments `1600 32 2 4`, 128 bits, about one minute); the margin table for the odd $m \leq 65$ in the proof of Proposition 7.1 (`margins_V.py`, Arb at 200 bits, which also lists every $m < 200$); the certificates of Theorem 9.2 (`dual_ref.py`, and `dualcert_arb2.py` with its library `dualcert.py`), the margins and the explicit numbers of Section 9 (`facts_M1.py`, Arb at 3000 bits). These are rigorous:

floating point is used only to choose the density and $\lambda$, and any choice gives a valid bound.

- *Not used by the proof of Theorem 1.1.* The direct certificates of Table 3 (`cert_V.py`, the program `cert_U.py` with only $z$ changed); the effective thresholds (`h0_U.py`, `h0_scan.py` at $m = 2$; `h0_V.py`, `h0_scan_V.py` at $m = 4$): exact sums, using the constants of Theorem 5.1; the identity at $m = 4$ in Section 8 (`id17_check.py`).
- *Checks of the lemmas* (not used by the proof). At $m = 2$: the machine and the matrix $\tilde{E}$ (`umach_core.py`); the bordered identity at $h = 2, \dots, 12$ and $X \in \{0, 1, -3/7\}$ (`bordered_check.py`); the row claim (`rowclaim_check.py`); Lemmas 3.2, 3.5 on exact profiles up to $h = 256$ (`law_check.py`, `u_profile.py` with `fastcontent.py`); Lemma 3.3 by brute force for every odd $d < K + 40$ and 194 values of $K$ (`law_check.py`); Lemma 4.3 entrywise for $h \le 64$ (`lemma10_check.py`); Lemma 6.1 (`blocks_check.py`, `lemmaP_check.py`). An AI referee reimplemented the machine without sharing code and reproduced the numbers it checked; the certificate reproduced bit for bit at $N = 100, 200, 1600$. For general $m$: Lemma 4.1 entrywise (`lemma10nu_check.py`), Lemmas 3.2, 3.5, 3.8 and 6.1 on exact data (`oddp_check.py`), and the scores of Table 10 (`vprofile.py`). The AI referee of Theorem 1.1 wrote an independent implementation of the machine for general $m$, which checked these lemmas in 47 pairs $(m, h)$, reproduced the scores it recomputed and the certificate at $m = 4$ bit for bit, and recomputed the margins independently.

The scripts are in the folder `scripts/paper` of the repository [17] (those for Section 9 in `scripts/paper/m1`), with a README that gives the command line for each of them and the small result files quoted here. Not included are the referees' own check scripts (apart from `dual_ref.py`) and the earlier script that produced the exact profiles for $h \le 128$ in Table 9; those profiles are included as data, and `u_profile.py` computes the same quantities (checked at $h = 16$). The scripts for general $m$ named above are in `scripts/paper/v`.

11.3. **Numerical evidence.** Table 9 shows exact scores at $m = 2$, and Table 10 at other $m$. The arithmetic column is $h^{-2} \sum_p n_p \log p$ for the exact contents, so the score is the sum of the two previous columns. The values approach the limits slowly, as $O(\log h / h)$.

11.4. **Controls.** No false instance of the method is available, since no unit with rational $\chi_2$ is known. As substitutes: the proved finite-$h$ bound of Remark 7.3 lies above the exact score at every tested $h$, and every per-prime inequality holds on the exact profiles, often with equality; for a unit of norm $+1$ the reflection breaks; with poles on one side only the conjugate place costs $+0.50$ per $h^2$ at $h = 16$; and at the points $\sqrt{2} - 1$ and $\varphi^{-1}$

| $h$ | score $h^{-2}\log\lvert F_h(T)\rvert$ | analytic | arithmetic | $n_2$ |
|---|---|---|---|---|
| 16 | $-0.2930$ | $-1.4379$ | 1.1449 | 54 |
| 32 | $-0.1734$ | $-1.4195$ | 1.2462 | 142 |
| 64 | $-0.1147$ | $-1.4015$ | 1.2869 | 350 |
| 96 | $-0.0951$ | $-1.3930$ | 1.2980 | 576 |
| 128 | $-0.0764$ | $-1.3876$ | 1.3112 | 830 |
| 192 | $-0.0659$ | $-1.3815$ | 1.3156 | 1344 |
| 256 | $-0.0596$ | $-1.3781$ | 1.3185 | 1918 |

TABLE 9. Exact scores for Corollary 1.2 ($m = 2$). At $h = 192$ and 256 the contents are proved lower bounds for each $n_p$ (fast content computation), which can only make the score look worse.

| $m$ | $h$ | score | analytic | arithmetic | $n_2$ |
|---|---|---|---|---|---|
| 4 | 16 | $-0.5479$ | $-1.7948$ | 1.2469 | 94 |
| 4 | 32 | $-0.3829$ | $-1.7739$ | 1.3910 | 349 |
| 4 | 48 | $-0.3502$ | $-1.7637$ | 1.4135 | 737 |
| 4 | 64 | $-0.3142$ | $-1.7572$ | 1.4429 | 1277 |
| 4 | 96 | $-0.2903$ | $-1.7491$ | 1.4587 | 2721 |
| 4 | 128 | $-0.2662$ | $-1.7443$ | 1.4781 | 4733 |
| 6 | 32 | $-0.5176$ | $-1.9798$ | 1.4622 | 142 |
| 6 | 64 | $-0.4434$ | $-1.9639$ | 1.5205 | 350 |
| 8 | 32 | $-0.5976$ | $-2.1254$ | 1.5278 | 558 |
| 8 | 64 | $-0.5096$ | $-2.1097$ | 1.6001 | 2205 |
| 11 | 32 | $-0.5330$ | $-2.2865$ | 1.7534 | 196 |
| 11 | 48 | $-0.4407$ | $-2.2769$ | 1.8362 | 401 |
| 3 | 32 | $-0.1326$ | $-1.6267$ | 1.4941 | 196 |
| 3 | 48 | $-0.0734$ | $-1.6165$ | 1.5431 | 401 |

TABLE 10. Exact scores at other $m$. The row $m = 3$ is in $\mathcal{E}$ and is not evidence for it (its score rises to $-0.0340$ at $h = 64$). At $m = 4$, $h = 96, 128$, the first computation gave proved lower bounds for the contents; an independent exact computation gave the same values.

the corresponding scores are positive (primal lower bounds $+0.3286$ and $+0.6572$ for the limiting exponent), so the method does not conclude there. At $\sqrt{2}-1 = z_1$ the exact scores are positive and rising (Section 7.2), and Lemma 6.1 fails at $\ell \mid m$ (Section 6), as it must. For Theorem 9.2, the method does not conclude at $n = 6$ ($M_1(6) \approx +0.477$), $n = 400$ ($M_1(400) \geq +0.070$) or $n = 720$ (numerical values).

## Acknowledgements and disclosure

The research, the construction, the proofs, the certificate programs, the Lean formalisation, the literature searches and the text of this paper were produced largely by AI models (Claude, Anthropic), working under the direction of the author. Apart from the check by the Lean kernel, all verification so far, including the internal reviews of the argument by AI referees, was carried out by AI systems. The proof of Theorem 1.1 for general $m$ was checked by one AI referee, at a single gate, with an independent implementation of the machine, and it is formalised in Lean (Section 10). No MathSciNet search or expert query has been made for it. The text of this revision, which adds Theorem 1.1, has not yet been reviewed as a whole. No human expert has yet reviewed the mathematics or the Lean code; we would be grateful for corrections. The prime number theorem in the formalisation is ported from the PrimeNumberTheoremAnd project of A. Kontorovich, T. Tao and collaborators [14], through the `zeta5` repository of César Del Solar [7] and the formalisation of [15].

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